The population status of birds in the United Kingdom, Channel Islands and Isle of Man: An analysis of conservation concern 2002-2007

Article *in* British Birds · September 2002

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The population status of birds in the United Kingdom, Channel Islands and Isle of Man:

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ABSTRACT This is the third assessment of the population status of birds in the UK. The listing criteria are based on global conservation status, historical population declines, recent population declines (both in numbers and in geographical range), European conservation status, breeding rarity, localised distribution, and the international importance of populations. For consistency, the criteria follow closely those used by previous reviews, with minor modifications where new information or methods are available. We have assessed the population status of 247 species in the UK: 40 (c. 16%) were placed on the 'Red list', 121 (c. 49%) on the 'Amber list', and 86 (c. 35%) on the 'Green list'. The number of Red-listed species has increased by four, and the number of Amber-listed species by eleven since the previous review in 1996. Nine new species have been added to the Red list because of declines in their breeding populations, while thirty-one species have remained Red listed. Five species have moved from the Red to the Amber list because their populations have more than doubled in the last 25 years.

Introduction

Red data lists for birds exist at a number of levels, from global (e.g. BirdLife International 2000) to national (e.g. Batten *et al.* 1990; Gibbons *et al.* 1996a; JNCC 1996), but have the shared aim of identifying population status changes and, where possible, focusing finite resources on the most pressing conservation priorities. The then Nature Conservancy Council and the RSPB published the first national Red list for British birds in 1990 (Batten *et al.* 1990). *Red Data Birds in Britain* used a series of largely quantitative criteria to judge the conservation status of each species. These criteria included the

258. Common Starling *Sturnus vulgaris.* A species which is new to the Red list, owing to the decline in its breeding population.

international importance of populations, the rarity of breeding species, population decline, localised distribution, and special concern. The list was subsequently updated and expanded, and published as *Birds of Conservation Concern* ('BoCC': Gibbons *et al*. 1996a) and *Birds of Conservation Importance* ('BoCI': JNCC 1996), which considered a wider range of qualifying criteria by virtue of improved data availability in both the UK and continental Europe. The revised listing considered population size and geographical range decline, historical population decline, rarity of breeding species, localised distribution, international importance, and both global and European conservation concern. Arguably, the most important advance in BoCC/BoCI, beyond the application of quantitative criteria, was the recognition that declines among widespread and common species should be reflected in the final listings. The main effect was to recognise the plight of a suite of species characteristic of farmland (see, for example, Aebischer *et al*. 2000).

We are fortunate that, in comparison with many other countries, the UK has a relatively long history of surveying and monitoring its birds. It is precisely because of this unparalleled volume and quality of data on population status and distribution that we are able to produce a list based on quantitative, rather than qualitative, criteria. Lists of population status need to be updated regularly to remain useful, and this paper provides a five-year update on BoCC/BoCI, as envisaged by Gibbons *et al*. (1996a) and JNCC (1996).

In producing a new list, the first task for the governmental and non-governmental organisations leading the project was to review the original listing criteria and consider alternative approaches. The most widely used guidelines of this kind are those published by the International Union for the Conservation of Nature (IUCN 2001) which quantify extinction probabilities and global conservation status for all taxa. We have used IUCN's official listings of global threats to birds (BirdLife International 2000) as an integral part of this review. The specific time frames, levels of population decline and thresholds for population size differ between the global criteria and those used in this paper because the IUCN criteria relate solely to extinction probability. In response to growing demand from individual countries, the IUCN is developing guidelines for the application of Red-list criteria at national and regional levels, although so far they have been applied in only a relatively small number of cases (Gärdenfors 2001; Gärdenfors *et al.* 2001; Keller

& Bollmann 2001).

A second approach, which has been adopted in North America, is the 'Partners in Flight Prioritisation Plan' (Beissinger *et al*. 2000; Carter *et al*. 2000). This system builds on earlier assessments of conservation concern known as 'Blue Lists' or 'Watch Lists', and sets out to apply quantitative and objective criteria to the available information. The criteria echo those used elsewhere and include geographical distribution, rarity, population trends, area importance, and perceived threats. There are many interesting features within the American system, such as the use of yearly rates of population change rather than trends over fixed periods, the use of statistical significance of these trends, and the consideration of species threat. Nonetheless, there appears to be no obvious advantage over the methods used by Gibbons *et al*. (1996a) and JNCC (1996), while there is much to be gained by maintaining continuity in the UK. In addition, while 'Partners in Flight' considers some aspects of international responsibility within the Americas, it does not consider them outside that area, and does not refer to the IUCN listings.

At the outset, it is important to distinguish between the assessment of extinction risk or population status (such as the IUCN and BoCC/BoCI lists) and any consequent setting of conservation priorities. The IUCN recommends that conservation priorities be set independently of population status and using additional criteria (Gärdenfors 2001; Gärdenfors *et al*. 2001). A priority list for conservation action should first take the biological information on population status and then consider additional factors, such as: the availability of funds or personnel to carry out actions; legal frameworks for the conservation of species; ecological, phylogenetic, historical, or cultural preferences for species; whether species are dependent on conservation action; expedience and opportunity for action; the likelihood of conservation success; and the degree to which species might act as flagships or be totemic of conservation action. It is important to note that, regarding this paper, any consequent setting of conservation priorities will need to take account of these factors.

In this paper, we review the population status of birds in the UK, the Channel Islands and the Isle of Man. We use seven quantitative criteria, based upon the previous assessments (Gibbons *et al*. 1996a; JNCC 1996), to assess the status of each species which occurs regularly, and place them on one of three lists. The simple categorisation of species as 'Red', 'Amber', or 'Green' gives one indication of the overall population status which, taken together with other information, should inform the setting of conservation priorities. Species which are 'Red listed' are those which are Globally Threatened, the population range of which has declined rapidly in recent years, or which

George McCarthy

259. Marsh Tit *Parus palustris*. A woodland species which is now Red-listed by virtue of its declining breeding population.

has declined historically and has not shown a recent substantial recovery. A species is 'Amber listed' if it has any of the following: an unfavourable population status in Europe; a very small population size; a population which has declined moderately over recent decades; distribution localised to a small number of sites; or if it occurs in internationally important numbers. In this way, the 'Red list' describes the population trends of the species, while the 'Amber list' describes both population status and UK international responsibility for populations.

One important function of the listings is to highlight the need for appropriate population monitoring. A monitoring framework is required which is able to detect future population changes in all species, although specific attention might be paid to comprehensive monitoring of Red-list species. Green-listed species will require a basic conservation provision (including site protection, where appropriate, and population monitoring). For example, Smew *Mergellus albellus*, and European Golden Plover *Pluvialis apricaria* are on Annex I of the EC Directive on the Conservation of Wild Birds, but Green listed by this review.

By following the methods of BoCC/BoCI, we can be confident that changes in the listings reflect true changes in population status and not changes in criteria (except where these are altered as a result of recent detailed examination of the data and consultation during the revision process). Others have adopted the methods we describe; for example, a review of population status in Ireland has been published recently (Newton *et al*. 1999).

In revising this list, we have been aware that the listing, when taken with the additional factors outlined above, might have important consequences for the priority accorded to monitoring requirements and conservation action for these species by government and non-governmental organisations. The Joint Nature Conservation Committee (JNCC) has adopted this objective review as one element of its ongoing *Species Status Assessment Programme*, which will in turn help to inform the revision of the UK Biodiversity Action Plan.

Species List

Our assessment covered all species on the most recent British Ornithologists' Union checklist (BOU 2001), excluding species which occur solely as vagrants (considered by the British Birds Rarities Committee), or scarce migrants (listed by Fraser *et al*. 1997, 1999, 2000), except those which, according to the RBBP, have bred; and species introduced by Man, either deliberately or accidentally, which have established self-sustaining breeding populations (listed as Category 'C' species by the BOU). Globally Threatened species (BirdLife International 2000) which have occurred in the UK at least once in each of the last 25 years were included in the analysis (Corn Crake *Crex crex*, Aquatic Warbler *Acrocephalus paludicola* and Scottish Crossbill *Loxia scotica)*. This criterion was designed to ensure that some of the world's most threatened species were evaluated, while excluding those species which, even if they occur occasionally as vagrants, could not be effectively targeted for conservation action in the UK.

The assessment is based on full species, rather than subspecies, races or distinct geographical populations. Where data are available for geographically discrete goose populations, however, we have made comparable assessments for these since they are often recognised for conservation action.

E. A. Janes

260. Stone-curlew *Burhinus oedicnemus*. This species remains Red-listed, because of a contraction in its breeding range, but is responding well to recent conservation action.

Methods

Listing criteria

The new list uses seven quantitative criteria to determine population status. These criteria assess the status of species based on global conservation status, recent population decline (both numbers and geographical range), historical population decline, European conservation status, breeding rarity, localised distribution, and international importance. 'Recent decline' is taken to refer to the last 25 years, and we distinguish between rapid (≥ 50% change over this period) and moderate (25-49.9%) declines in numbers or range to define Red- and Amber-list species respectively. For a small number of species, information on population trends is only available for periods shorter or longer than 25 years and we retain the same numerical thresholds for these, for the sake of simplicity and the greater importance attached to the magnitude of the decline.

Global conservation status (IUCN)

Qualification *IUCN Globally Threatened: Critically Endangered, Endangered or Vulnerable = Red list*

This criterion assesses the conservation status of each species at a global scale. In essence, it puts the UK status of a species into a global context and has been used as an independent rationale for judging national status. We have used the most up-to-date assessment of Globally Threatened species, as published in *Threatened Birds of the World* (BirdLife International 2000), to identify UK species of global conservation concern.

Recent Population Decline

Decline in breeding population (BDp, BDMp)

Qualification *Population decline ≥ 50% over 25 years = Red list; 25-49.9% over 25 years = Amber list*

To determine the population trend for each breeding species, we used data from a variety of survey and monitoring schemes. These included the BTO/JNCC Common Birds Census (CBC) and Waterways Bird Survey (WBS), the BTO/RSPB/JNCC Breeding Bird Survey (BBS), the JNCC/RSPB/SOTEAG Seabird Monitoring Programme and Seabird 2000, data from the Rare Breeding Birds Panel (RBBP), and single-species surveys, mostly conducted under the Statutory Conservation Agencies and RSPB Annual Breeding Bird Scheme (SCARABBS) Agreement. These data were used in the following manner to assess whether species qualified as declining breeders.

Common Birds Census (CBC)

Population trends using CBC indices were based on data from all CBC plots combined – 'farmland', 'woodland' and 'special' (mainly heathland and wetland) – between 1966 and 2000. Population changes were estimated using a Generalised Additive Model (GAM), a type of log-linear regression model which incorporates a smoothing function (Fewster *et al.* 2000). Counts were modelled as the product of site and year effects, on the assumption that between-year changes were homogenous across plots. 'Smoothing' was used to remove the effects of short-term fluctuations (e.g. those caused by periods of severe weather) in order to reveal the underlying pattern of population change. In this analytical method, the endpoints of the derived smoothed plots are excluded from the trends. Thus, although the full run of data from 1966-2000 is required to derive the annual indices, the population trends cover the 25-year period 1974-1999. These methods represent a considerable refinement over the quadratic regressions used by Gibbons *et al.* (1996a), and some of the changes may reflect more appropriate and accurate models.

We have used the population trends for all species routinely monitored by the CBC, although we have been conscious of the uneven geographical distribution of sampling plots in the UK. For some species, the CBC may provide only a small or unrepresentative sample of its national population and this will affect the level of precision and bias of the population trends. Those species trends that were simply based on small sample sizes have been used without qualification. This is because there is no reason to believe that they are wrong (i.e. biased), even though they may be imprecise because of low samples. The same may not be true, however, where the CBC monitors an unrepresentative part of the species population. To determine whether this was the case, we have used the measures of abundance from the *New Atlas* (Gibbons *et al.* 1993) to assess how representative CBC trends were. The ratio of the mean abundance over all 10-km squares, to the mean abundance in 10-km squares containing CBC plots, was calculated for each species (see table

7, 'correction ratio'; Gibbons *et al*. 1993). A species whose correction ratio was greater than or equal to 1.0 (e.g. Meadow Pipit *Anthus pratensis*) occurred mainly in areas without CBC plots, and so the CBC would be unlikely to represent national trends in population levels. CBC trends *were* likely to be representative for species whose correction ratio was less than 1.0 (i.e. CBC plots encompassed the species' range well). In our assessment, species for which the CBC trend was unrepresentative have not been admitted to the Red list, even if the CBC indicated a decline of more than 50% over the 25-year period. This is because there are good reasons to believe the trend may be a biased representation of its true, but unmeasured, UK trend. Because the true rate of decline could be less, similar to, or more severe than that measured, however, these species have been admitted to the Amber list as a precautionary measure.

We could have followed the general principle of placing species with unrepresentative declines of more than 50% on the Amber list rather than on the Red list, and by placing all those with unrepresentative declines of 25-49.9% on the Green list. We have not done this because we feel that these declines might, at least in part, be real. If such species were Green listed, their true status may be overlooked for some time to come.

Waterways Bird Survey (WBS)

Population trends from WBS indices were based on data from all plots between 1974-2000. The data were analysed in the same manner as for CBC data, using a GAM with a smoothing function which removes short-term fluctuations to reveal the underlying pattern of population change. As for the CBC data, the endpoints of the smoothed plots were excluded, so the change in population indices is for the 24-year period 1975-99.

Breeding Bird Survey (BBS)

BBS data were examined because they have the potential to identify important population trends for species which are poorly covered by the CBC, even though they represent only a short run of data. Population trends using BBS indices were based on data for the six-year period 1994-2000 (the full dataset available). Counts of each species were summarised for each 1-km square surveyed and then analysed using a log-linear regression model with full year and site effects to estimate annual indices (Noble *et al*. 2000). The trends were not smoothed because this was considered inappropriate given the short run of data. These indices incorporate a weighting factor to correct for differences in survey coverage between regions. The extent of population change was calculated for each species as the percentage change between the index for 1994 and 2000.

The population trends of species that were based on data from the CBC, WBS or BBS were used in a hierarchical manner. We first used the CBC trends to assign listings, except for four riparian species the populations of which were not well monitored by the CBC and the listings of which were based solely on their WBS trend (Common Sandpiper *Actitis hypoleucos*, Common Kingfisher *Alcedo atthis*, Grey Wagtail *Motacilla cinerea* and Dipper *Cinclus cinclus*). Where the CBC trends were subject to data caveats (i.e. unrepresentative samples), the trends based on WBS or BBS were explored. If either of the latter trends were judged more appropriate (in terms of being representative), these were then used in our assessment; otherwise, the CBC trend was used. These choices were led by data quality and not by a precautionary approach, which in this instance might alter the status of species incorrectly.

Alternative data sources were investigated for five CBC species whose trends might not be representative. For example, Little Grebe *Tachybaptus ruficollis* showed a 2% increase on CBC plots and a 57% decline on WBS plots over roughly the same period. In this case, both sources are biased towards particular habitat types, and the CBC trend was retained in the assessment. Common Snipe *Gallinago gallinago*, Woodcock *Scolopax rusticola* and Common Redshank *Tringa totanus* show very strong declines on CBC plots, but they have been Amber listed on the basis of these data because of small sample sizes in recent years and because of particular concerns over habitat and geographical representation (table 7). In this case, breeding Common Redshanks are found mainly on saltmarsh and significant populations of Woodcock and Common Snipe breed in northern Britain, which are habitats and areas not well covered by the CBC. For Eurasian Curlew *Numenius arquata*, the CBC trend suggests a moderate decline, whereas the WBS, having a more northerly distribution, which is arguably more representative of its breeding habitat, shows strong gains. We have, therefore, used the WBS trend in our assessment.

Seabird trends

Breeding seabirds in the UK are monitored by two closely related schemes. The first is a census of all breeding seabirds, undertaken approximately every 15 years: 1969-70 (Cramp *et al.* 1974), 1985-88 (Lloyd *et al.* 1991) and 1998-2001 (Mitchell *et al.* in prep.). Population changes were assessed over the longest period for which comparable data were available, i.e. either from 1969/70 to 1998/01, or from 1985/88 to 1998/01. The second scheme, the Seabird Monitoring Programme, was instigated in 1986 to measure change at a sample of colonies throughout the UK on an annual basis (Reid 2000; Tasker 2000). Trend information was drawn from the most appropriate source, or sources.

Rare Breeding Birds Panel (RBBP)

Population trends using RBBP data were based on data for 1973-99: the period over which the Panel has operated and published its results. Trends were calculated for each of the species monitored by the Panel for which there was at least one confirmed breeding pair (where 'breeding' is defined as, at least, the laying of presumed fertile eggs) during the most recent five-year period 1995-99 (i.e. species qualifying under the breeding rarity category, see below). This excludes transient breeders, i.e. species nesting very irregularly and with many years between nesting attempts (e.g. Black-winged Stilt *Himantopus himantopus*, Hoopoe *Upupa epops*), which are not an established part of the UK's breeding avifauna. Means of the maximum total number of pairs (or occasionally more relevant units; see table 9) were calculated for the 1973-77 and 1995-99 five-year periods. Population change was then calculated as the percentage change between the two five-year means. Five-year means have been used because they smooth the inter-annual variation in numbers, thus making the percentage change estimates more reliable. Inevitably, this reduces the mean time interval over which the population trend is measured to 22 years, although data from 27 years, 1973 to 1999, were used.

Single-species surveys

Many species are too rare to be reliably monitored by the CBC, WBS and BBS, yet are too common to be covered by the RBBP. For many of these species (e.g. Grey Heron *Ardea cinerea*), customised single-species surveys have been undertaken, often repeated at varying time intervals. Population trends based on data from comparable surveys of single species were calculated over the longest period available, since many of these species lack 25-year trends. For Stone-curlew *Burhinus oedicnemus*, the population trend covered the 30-year period 1970-2000, with the estimate of population size from the first breeding atlas (Sharrock 1976) providing the 1970 estimate, and an annual census the 2000 estimate. For this species, although a 30-year trend was available, a 25-year trend was not. Where a species was surveyed in only part of its UK range, population change was assessed using data from only those geographical areas covered by both surveys (e.g. Red-throated Diver *Gavia stellata* in Shetland). The degree of population change was calculated for each species as the percentage change in the population estimates between the two surveys.

A small number of species lack either co-ordinated UK-wide, or comparable survey data. For two such species, Northern Lapwing *Vanellus vanellus* and Ring Ouzel *Turdus torquatus*, we have used a variety of population data to assess population change.

Decline in breeding range (BDr, BDMr)

<u>Qualification</u> Range decline ≥ 50% over 25 years = Red list; 25-49.9% over 25 years = Amber list

To assess changes in geographical range, we used data from the two atlases of breeding birds in Britain & Ireland (Sharrock 1976; Gibbons *et al.* 1993). Although now ten years old, Gibbons *et al.* (1993) provides the only recent data on geographical range for the majority of species breeding in the UK. Where available, we have used more recent complete census data to assess distributional change over a longer period. In addition, comprehensive surveys are carried out annually for some species (e.g. Slavonian Grebe *Podiceps auritus* and Great Bittern *Botaurus stellaris*), and we have used the most recent data available. For those species that were not surveyed throughout their entire UK range (e.g. Corn Crake), the assessment used data from only those geographical areas covered during both survey periods. Comparison of bird distributions assessed 20 years apart using slightly different methods is problematic. Overall, survey coverage during the two atlases was comparable in Britain as a whole, although there was regional variation (Gibbons *et al.* 1993). Hence, distributional change was assessed as follows. Changes in range using data from the two breeding atlases cover the 20-24-year period between 1968-72 and 1988-91. For each species, we calculated the percentage change in the number of occupied 10-km squares in the UK (breeding 'confirmed', 'probable' or 'possible' for the 1968-72 *Atlas* and 'breeding' or 'seen' for the 1988-91 *New Atlas*). We excluded those species which occurred in 20 or fewer 10-km squares during both *Atlas* periods, because the percentage change calculations can lead to artificially high or low values. In calculating the range change, we used only those 10-km squares that were covered during both periods. In addition, data from each of the Channel Islands (Jersey, Guernsey, Alderney, Sark and Herm) and from Fair Isle, Shetland, from the 1988-91 *New Atlas* were each treated as single 10-km squares, even though in total they actually cover 18 10-km squares, in order to allow comparison with the 1968-72 *Atlas* (see p. 18 of Gibbons *et al.* 1993 for more details). Because seabirds frequently occur away from their breeding colonies, only 10-km squares with evidence of breeding ('probable' or 'confirmed' in 1968-72, and 'breeding' in 1988-91) were used to assess changes in distribution for these species.

More recent complete census data (i.e. covering all of the breeding range) were available for 13 species (Slavonian Grebe, Great Bittern, Common Scoter *Melanitta nigra*, Spotted Crake *Porzana porzana*, Corn Crake, Stone-curlew, Whimbrel *Numenius phaeopus*, Roseate Tern *Sterna dougallii*, European Nightjar *Caprimulgus europaeus*, Wood Lark *Lullula arborea*, Rufous Nightingale *Luscinia megarhynchos*, Crested Tit *Parus cristatus* and Cirl Bunting *Emberiza cirlus*). Changes in distribution were calculated for each of these species in a similar manner to that outlined above, using data from the 1968-72 *Atlas* as the baseline and comparing the number of occupied 10-km squares ('possible', 'probable' or 'confirmed') for both time periods. As for other seabirds, for Roseate Tern only 10-km squares with evidence of breeding ('probable' or 'confirmed' for both time periods) were used. Similarly, range change for Corn Crake has been based on records of 10-km squares with evidence of breeding only (see, for example, Green & Gibbons 2000).

As the estimates of geographical distribution from the 1968-72 *Atlas* are an accumulated pattern over a five-year period, the documented range of some species may have been greater than would have been recorded by a census in a single year. Range loss may thus be exaggerated when comparing the *Atlas* with a recent census undertaken in a single year. Conversely, all recent single-species surveys have been carried out by surveyors dedicated to that species alone, while fieldworkers for the 1968-72 *Atlas* were expected to record all species in their grid square. In general, we are unable to take account of these changes in survey effort, although Green & Gibbons (2000) have assessed

Chris Gomersall

261. Black Grouse *Tetrao tetrix* remains on the Red list.

the extent of this potential bias by estimating the Corn Crake's range in a single year around 1970. This was then compared with the most recent complete census in 1998. Unfortunately, we have been unable to perform such analyses for the remaining 12 species.

Decline in population during the non-breeding season (WDp, WDMp)

Qualification Population decline ≥ 50% over 25 years = Red list; 25-49.9% over 25 years = Amber list

To assess the population trends of species during the non-breeding season, we used data from the Wetland Bird Survey (WeBS) and co-ordinated Wildfowl & Wetlands Trust/JNCC goose counts. Data are available for a number of waterbirds, principally wildfowl and waders. Population-trend data are lacking for most terrestrial species outside the breeding season, although most species that winter in the UK also breed here.

Population trends using WeBS indices were based on data for the 25-year period from winter 1974/75 to winter 1999/2000. The WeBS count data were smoothed using a GAM to remove year-to-year variation and so reveal the underlying population trend (Atkinson & Rehfisch 2000). The level of population change was calculated for each species as the percentage change in the smoothed index between 1974/75 and 1999/2000.

Population trends of Pink-footed Geese *Anser brachyrhynchus*, Greylag Geese *Anser anser* from Iceland, Barnacle Geese *Branta leucopsis* from both Svalbard and Greenland, and light-bellied Brent Geese *Branta bernicla hrota* from Svalbard, none of which are adequately monitored by WeBS, were based on data from co-ordinated goose counts. Trends were based on the maximum number of birds counted because these regular censuses record almost the entire population during the non-breeding season in the UK. For these species, population change was calculated as the percentage change in the total count between 1974 and 1999. For Greenland White-fronted Geese *Anser albifrons flavirostris* and East Canadian light-bellied Brent Geese, the population trends were based on data for the period between 1986/87 and 1999/2000 and the period from 1982/83 to 1999/2000 respectively, the periods during which reliable monitoring data were available.

Population trends were not calculated for several species monitored by WeBS (divers, grebes, Great Cormorant *Phalacrocorax carbo*, herons, seaducks, rails, waders, gulls, terns and Common Kingfisher) because it is not known whether the counts are representative of national numbers.

Roger Wilmshurst

262. Red Kite *Milvus milvus*. A good example of a conservation success story; Red Kite was formerly Red-listed, but is now classified as an Amber list species.

Historical decline in breeding population (HD, HDrec)

Qualification Severe historical population decline during 1800-1995 = Red list

The time period of 25 years is relatively short for assessing population trends, and is designed to capture current and ongoing conservation issues. This is also partly based on pragmatism because few major monitoring schemes existed before the 1960s. It would thus be possible for a species to have undergone a large population decline over the last two centuries, but for its population to have remained stable over the last two decades. Such a species would not qualify as a declining breeding species and would not be considered a species of concern, even though current population levels are very much lower than 200 years ago. To assess historical population trends for each species, we used the assessment of status given by Gibbons *et al.* (1996b). Building upon earlier reviews, the population trend for each species, in each of five-time periods during 1800-1995, was assessed using an ordinal system, from -5 ('huge and widespread decline') to +5 ('spectacular increase'), following Alexander & Lack (1944), and Parslow (1973) (see table 1; Gibbons *et al.* 1996b). The overall population trend during 1800-1995 was calculated as the sum of scores across all five-time periods. Although this system involved a semi-quantification of qualitative assessments, it allowed a comparison of trends between species and, importantly, it was possible to determine those species the breeding populations of which had declined most since 1800. Those extant species whose overall population trend was less than or equal to -10 qualified as historically declining species in both the original BoCC/BoCI listings, and in this review (though with a few exceptions, outlined below).

Given the manner in which this historical trend was calculated, it is difficult for any species to leave this category, and hence the list, for many decades. Thus, we have introduced the following sub-criteria to identify species that should, by virtue of more recent population trends, move from Red to Amber.

Species have been moved from Red to Amber if their population size has more than doubled in the last 25 years, providing that they had met a minimum threshold (in this review we have taken this to be 100 breeding pairs), and were not classified by IUCN as Globally Threatened.

In future revisions, we recommend that species could be moved from Amber to Green if they do not qualify for listing under any other criteria (e.g. localised or rare breeder), and their population continues to recover (i.e. numbers increase by ≥20% between this and the next review).

A species which had declined historically would move back on to the Red list if its modern population recovery falters (i.e. numbers decrease by ≥20% between this and the next review).

These sub-criteria for moving species off the list of historically declining species are designed to recognise conservation success, while not jeopardising the status of fragile populations or ignoring international responsibilities.

European conservation status (SPEC 2 & 3)

Qualification SPEC 2 & 3 = at least Amber list

This criterion assesses the conservation status of each species in a European context. The conservation status of all European birds was assessed in *Birds in Europe* (Tucker & Heath 1994; this assessment is currently being revised). Four categories of conservation concern were recognised and are known as Species of European Conservation Concern (SPECs). SPEC1 species are of global conservation concern, SPEC2 species are of unfavourable status and concentrated in Europe (more than 50% of the world population or range in Europe), and SPEC3 species are of unfavourable status but not concentrated in Europe. The criteria for SPEC status include rarity, localisation and decline in Europe, and can apply to both breeding and non-breeding populations (Tucker & Heath 1994). Species classed as either SPEC2 or SPEC3 have been placed on the Amber list, except for four species (Smew, Jack Snipe *Lymnocryptes minimus*, Little Gull *Larus minutus*, and Black Tern *Chlidonias niger*), which occur only as non-breeding species in the UK.

Breeding rarity (BR)

Qualification less than 300 pairs = at least Amber list

To determine which species qualified as rare breeders, we used data from Rare Breeding Birds Panel (RBBP) reports (Ogilvie *et al.* 1998, 1999a & b, 2000, 2001), from single-species surveys and from Seabird 2000 data. Only those species reported by the RBBP for which there was at least one confirmed breeding pair (see above) during the five-year period 1995-99 were assessed. Population size was calculated for each as the mean of the maximum total number of pairs (or occasionally more relevant units; see table 9) during the five-year period 1995-99; the latter being the most recent year of reporting available. Species qualified under this criterion with a mean of 1-300 breeding pairs. The use of a five-year running mean avoids having continually to rewrite the list following annual changes in population levels, yet enables the list to be sufficiently responsive to change. Inevitably, it means that rare species with increasing populations will remain as rare breeders even though their populations have passed 300 pairs in the most recent year or years. This may, however, be advisable since it is a precautionary approach. Recent information was available for a small number of species from intensive or systematic surveys. For these species (e.g. Red Kite *Milvus milvus*, Roseate Tern, and Cirl Bunting), we used the estimates of UK population size derived from these surveys, rather than the five-year means based on RBBP data to assess whether they qualified as rare breeders.

Localised breeding (BL) and non-breeding species (WL)

<u>Qualification</u> ≥ 50% of the breeding or non-breeding population occurs at ten or fewer sites = at least Amber list

This criterion was used because a species with a population confined to a small number of sites faces a greater threat from chance events (either natural or anthropogenic) than a more widespread species. Rare breeders were not considered as localised breeding species because their populations are, generally, highly localised anyway.

We undertook two separate analyses to determine how localised each species was during the breeding and non-breeding seasons. Both were aimed at determining what proportion of the UK population occurs at the ten best sites for each species, in each season. In the first analysis, we defined each of these sites as an Important Bird Area (IBA) (Heath & Evans 2000). In the second, we defined each site as a Special Protection Area (SPA) (Stroud *et al.* 2001).

In the first analysis, the sum of counts at the ten most important IBAs for each species was calculated, for each relevant season (breeding and non-breeding), and then divided by the estimated UK population size, again for the relevant season. We also assessed the counts of species on passage for the small number of wader species with UK population estimates. We ignored the counts of non-breeding flocks during the breeding season in this analysis, however, because UK population estimates of this component of the population are lacking. This process was then repeated using SPA data. A species qualified as localised when the resulting value was greater than or equal to 0.5 for IBAs and/or SPAs. We have taken care to use the most recent estimates of UK population size for each species in each relevant season (BirdLife International/European Bird Census Council 2000; Stone *et al.* 1997), but updated for wildfowl (Kershaw & Cranswick in prep. for Britain, and 5-year-mean peak WeBS counts for Northern Ireland) and waders (Rehfisch *et al.* in prep.). For some species, the UK population estimates were presented as ranges, and so it was possible to calculate a minimum and a maximum value for the percentage of the UK population occurring on the ten best sites. Only if both the maximum and minimum values were greater than or equal to 0.5, did the species qualify for this criterion.

Each of these two approaches, one using IBA data, the other using SPA data, has advantages and disadvantages. The IBA data include counts of all species (with the exception of common and widespread ones) on all sites, yet the data are somewhat out of date, mostly from the early to mid 1990s. Some IBA 'sites' are also extremely large (e.g. more than 230,000 ha). The SPA data are slightly more up to date, but cover only those species for which the site has been designated as an SPA. Species could qualify based on either IBA or SPA data because both are comprehensive national assessments of networks of important sites.

International importance during the breeding (BI) or non-breeding (WI) season

<u>Qualification</u> ≥ 20% of the European breeding or non-breeding population occurs in the UK = at least Amber list

Breeding species are considered to occur in internationally important numbers if 20% or more of the European population breeds in the UK. To put this figure into context, the UK is about 2.5% of the European land mass. To assess whether species occur in internationally important numbers during the breeding season, we used the estimates of European and UK breeding population sizes from *European bird populations: estimates and trends* (BirdLife International/European Bird Census Council 2000). Where more up-to-date estimates are available for the UK, we have used these, and adjusted the European population estimates accordingly. Using these data, we calculated the proportion of the European population that breeds in the UK for each species. For some species, the European (and in some cases the UK) breeding population estimates were presented as maximum and minimum values. Only if both the maximum and minimum values of the percentage of the European population breeding in the UK were greater than or equal to 20% did species qualify under this criterion.

Unfortunately, the European population size of many species during the non-breeding season is not well known, though data are available for waterbirds. Non-breeding waterbirds are considered to occur in internationally important numbers if 20% or more of the northwest European or East Atlantic Flyway population occurs in the UK. We calculated non-breeding waterbird population sizes for the UK from two sources of data. Recently revised estimates have been produced for wildfowl in Britain (Kershaw & Cranswick in prep.) and waders in the UK (Rehfisch *et al.* in prep). For wildfowl in Northern Ireland, population estimates were based on WeBS data and calculated as the mean annual maxima over the five-year period from 1994/95 to 1998/99. An approximation of the UK non-breeding population was then calculated by summing the British and Northern Ireland population estimates. Population sizes of non-breeding waterbirds for northwest Europe and the East Atlantic Flyway were also taken from Rose & Scott (1997). It should, however, be noted that many of the original data in that source, and especially those for waders, come from the early to mid 1980s (e.g. Smit & Piersma 1989). Where population estimates were given as ranges, the maximum value was used. For eight species, the estimates are given as 'greater than one million', in which case one million was used in the assessment. In no case do the UK estimates come close to the threshold figure of 20%, however, and of course if the European estimates were larger, the UK fraction would be even smaller. Hence, this approach is conservative.

Results

We have assessed the population status of 247 species. Forty species (*c.*16%) were placed on the Red list and 121 species (*c.*49%) on the Amber list. The species that qualified as Red or Amber listed, and the criteria under which they qualify, are given in tables 1 & 2. The remaining 86 species (*c.*35%) were Green listed by default, although in a number of cases this probably reflects a lack of data rather than population status (table 3). This is a particular issue for species occurring on passage in the UK. A small number of previously regular breeding species have become extinct as breeding birds in the UK since 1800 (table 4).

Nine new species have been added to the original BoCC Red list, because of declines in breeding populations (table 5). Of these, six have moved from Amber to Red (Ring Ouzel, Grasshopper Warbler *Locustella naevia*, Savi's Warbler *L. luscinioides*, Marsh Tit *Parus palustris*, Willow Tit *P. montanus* and Common Starling *Sturnus vulgaris*), and three species (Lesser Spotted Woodpecker *Dendrocopos minor*, House Sparrow *Passer domesticus*, and Yellowhammer *Emberiza citrinella*) have moved straight from Green to Red. Lesser Spotted Woodpecker and Grasshopper Warbler qualify because this review has adopted the precautionary approach, so that species qualify when the data are considered representative, even when the trends are based on small sample sizes. This was not the case in the original review, where trends based on low numbers of CBC plots were excluded (Gibbons *et al.* 1996a). Despite this, the Lesser Spotted Woodpecker's trend would not have qualified it for Red or Amber listing at that time. Ring Ouzel qualifies because its population, when corrected for differences in survey effort between the 1988-91 *New Atlas* and the 1999 national survey, shows a decline of more than 50% over ten years. The population of Savi's Warblers, although small, has undergone a marked decline over the last 25 years (table 7). Note that some of the species added to the Red list still have populations in excess of 10,000 pairs/territories in the UK. While still relatively common, the status of such species, showing considerable recent declines, is of general concern.

Five species originally listed under the historical decline category have moved from the

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Table 1. The population status of birds in the UK: 'Red-listed' species.

Species	Scientific name	Red-listing criteria					Additional Amber-listing criteria under which species qualifies									
		IUCN	HD	BD p	BD r	WD p	HD rec	BDM p	BDM r	WDM p	SPEC 2 & 3	BR	BL	WL	BI	WI
Great Bittern[1]	*Botaurus stellaris*		*	*	*						3	*		*		
Common Scoter	*Melanitta nigra*				*			*				*		*		
White-tailed Eagle[1]	*Haliaeetus albicilla*		*								3	*				
Hen Harrier[1]	*Circus cyaneus*		*								3					
Black Grouse	*Tetrao tetrix*		*	*					*		3					
Capercaillie[1]	*Tetrao urogallus*			*	*											
Grey Partridge[2]	*Perdix perdix*			*							3					
Common Quail	*Coturnix coturnix*		*								3					
Corn Crake[1]	*Crex crex*	A	*		*								*			
Stone-curlew[1]	*Burhinus oedicnemus*				*						3	*				
Black-tailed Godwit	*Limosa limosa*		*					*			2	*		*p		*
Red-necked Phalarope[1]	*Phalaropus lobatus*		*									*				
Roseate Tern[1]	*Sterna dougallii*			*	*						3	*				
Turtle Dove[2]	*Streptopelia turtur*			*							3					
European Nightjar[1]	*Caprimulgus europaeus*				*						2		*			
Wryneck	*Jynx torquilla*		*	*	*						3	*				
Lesser Spotted Woodpecker	*Dendrocopos minor*			*												
Wood Lark[1]	*Lullula arborea*				*						2		*			
Sky Lark[2]	*Alauda arvensis*			*							3					
Ring Ouzel	*Turdus torquatus*			*					*							
Song Thrush[2]	*Turdus philomelos*			*												
Grasshopper Warbler[2]	*Locustella naevia*			*					*							
Savi's Warbler	*Locustella luscinioides*			*								*				
Aquatic Warbler[1]	*Acrocephalus paludicola*	B												p		
Marsh Warbler	*Acrocephalus palustris*			*					*			*				
Spotted Flycatcher[2]	*Muscicapa striata*			*							3					
Marsh Tit[2]	*Parus palustris*			*												
Willow Tit[2]	*Parus montanus*			*												
Red-backed Shrike[1]	*Lanius collurio*		*	*	*						3	*				
Common Starling[2]	*Sturnus vulgaris*			*												
House Sparrow[2]	*Passer domesticus*			*												
Tree Sparrow[2]	*Passer montanus*			*												
Linnet[2]	*Carduelis cannabina*			*												

Twite	*Carduelis flavirostris*		*				
Scottish Crossbill[1]	*Loxia scotica*	C					*
Bullfinch[2]	*Pyrrhula pyrrhula*			*			
Yellowhammer[2]	*Emberiza citrinella*			*			
Cirl Bunting	*Emberiza cirlus*				*		
Reed Bunting[2]	*Emberiza schoeniclus*			*			
Corn Bunting[2]	*Miliaria calandra*		*	*		*	

KEY
- * The species qualifies for Red or Amber listing under a particular criterion.
- 1 Annex I of the EC Directive on the Conservation of Wild Birds.
- 2 Estimated population size of >10,000 pairs or territories in the UK.

Qualifying criteria for:

Red listing

IUCN	Globally Threatened
HD	Historical population decline during 1800-1995
BDp	Rapid (≥ 50%) decline in UK breeding population over previous 25 years
BDr	Rapid (≥ 50%) contraction of UK breeding range over previous 25 years
WDp	Rapid (≥ 50%) decline in UK non-breeding population over previous 25 years

Amber listing

HDrec	Historical population decline during 1800-1995, but recovering; population size has more than doubled over previous 25 years
BDMp	Moderate (25-49%) decline in UK breeding population over previous 25 years
BDMr	Moderate (25-49%) contraction of UK breeding range over 25 years
WDMp	Moderate (25-49%) decline in UK non-breeding population over previous 25 years
SPECs 2 & 3	Species with unfavourable conservation status, concentrated in Europe (2) or not (3). W = wintering population only
BR	5-year mean of 1-300 breeding pairs in UK
BL	≥ 50% of UK breeding population in ten or fewer sites, but not BR
WL	≥ 50% of UK non-breeding population in ten or fewer sites; p = passage population
BI	≥ 20% of European breeding population in UK
WI	≥ 20% of NW European (wildfowl), East Atlantic Flyway (waders), or European (others) non-breeding populations in UK

IUCN Globally Threatened species assessment (BirdLife International 2000):

A Vulnerable A2c (= Decline >20% in 10 years or 3 generations; decline likely in near future (projected or suspected) based on: decline in extent of occurrence, area of occupancy and/or quality of habitat)

B Vulnerable A1c, A2c (= Decline >20% in 10 years or 3 generations; decline which has happened (observed, estimated, inferred or suspected), and decline likely in near future (projected or suspected) based on: decline in extent of occurrence, area of occupancy and/or quality of habitat)

C Vulnerable B1, B3a, B3c (Extent of occurrence <20,000km^2, severe fragmentation; extreme fluctuations in extent of occurrence and number of locations [R. Summers pers com.]).

Table 2. The population status of birds in the UK: 'Amber listed' species.

Species	Scientific name	Qualifying criteria for Amber listing									
		HD rec	BDM p	BDM r	WDM p	SPEC 2&3	BR	BL	WL	BI	WI
Red-throated Diver[1]	*Gavia stellata*		*			3					
Black-throated Diver[1]	*Gavia arctica*					3	*				
Great Northern Diver[1]	*Gavia immer*										*
Red-necked Grebe	*Podiceps grisegena*						*		*		
Slavonian Grebe[1]	*Podiceps auritus*		*				*				
Black-neckedGrebe	*Podiceps nigricollis*						*				
Fulmar	*Fulmarus glacialis*							*			
Manx Shearwater	*Puffinus puffinus*					2		*		*	
European Storm-petrel[1]	*Hydrobates pelagicus*					2		*			
Leach's Storm-petrel[1]	*Oceanodroma leucorhoa*					3		*		*	
Northern Gannet	*Morus bassanus*					2		*		*	
Great Cormorant	*Phalacrocorax carbo*							*			*
Shag	*Phalacrocorax aristotelis*									*	
Little Egret[1]	*Egretta garzetta*						*				
EurasianSpoonbill[1]	*Platalea leucorodia*					2	*				
Mute Swan	*Cygnus olor*									*	
Tundra Swan[1]	*Cygnus columbianus*					3w			*		*
Whooper Swan[1]	*Cygnus cygnus*						*		*		*
Bean Goose	*Anser fabalis*								*		
Pink-footed Goose	*Anser brachyrhynchus*								*		*
White-fronted Goose	*Anser albifrons*				a				*		b
Greylag Goose	*Anser anser*							*	*		*
Barnacle Goose[1]	*Branta leucopsis*					2w			*		*
Brent Goose	*Branta bernicla*					3w			*		*
Common Shelduck	*Tadorna tadorna*								*		*
Eurasian Wigeon	*Anas penelope*								*		*
Gadwall	*Anas strepera*					3		*			*
Eurasian Teal	*Anas crecca*										*
Pintail	*Anas acuta*					3	*		*		*
Garganey	*Anas querquedula*					3	*				
Northern Shoveler	*Anas clypeata*										*
Common Pochard	*Aythya ferina*										*
Greater Scaup	*Aythya marila*					3w	*		*		
Common Eider	*Somateria mollissima*								*		
Long-tailed Duck	*Clangula hyemalis*								*		
VelvetScoter	*Melanitta fusca*					3w			*		
Common Goldeneye	*Bucephala clangula*						*		*		
European Honey-buzzard[1]	*Pernis apivorus*						*				
Red Kite[1]	*Milvus milvus*	*									
Marsh Harrier[1]	*Circus aeruginosus*	*					*				
Montagu's Harrier[1]	*Circus pygargus*			*			*				
Golden Eagle[1]	*Aquila chrysaetos*					3					
Osprey[1]	*Pandion haliaetus*	*				3	*				
Common Kestrel	*Falco tinnunculus*		*			3					
Merlin[1]	*Falco columbarius*	*									
Peregrine Falcon[1]	*Falco peregrinus*					3					
Red Grouse	*Lagopus lagopus*		*								
Water Rail	*Rallus aquaticus*			*							
Spotted Crake[1]	*Porzana porzana*						*				
Common Crane	*Grus grus*					3	*				
Oystercatcher	*Haematopus ostralegus*								*	*	*
Avocet[1]	*Recurvirostra avosetta*					3w		*	*p		
Great Ringed Plover	*Charadrius hiaticula*				*						*
Dotterel[1]	*Charadrius morinellus*							*			
Grey Plover	*Pluvialis squatarola*								*p		*
Northern Lapwing	*Vanellus vanellus*		*								*
Red Knot	*Calidris canutus*					3w			*		*
Temminck's Stint	*Calidris temminckii*						*				
Purple Sandpiper	*Calidris maritima*						*				*
Dunlin	*Calidris alpina*				*	3w		*	*		*
Ruff[1]	*Philomachus pugnax*						*		*		
Common Snipe	*Gallinago gallinago*		*c								
Woodcock	*Scolopax rusticola*		*c	*		3w					
Bar-tailed Godwit[1]	*Limosa lapponica*					3w			*p		*

Species	Scientific name	HD rec	BDM p	BDM r	WDM p	SPEC 2&3	BR	BL	WL	BI	WI
Whimbrel	*Numenius phaeopus*							*	p		
Eurasian Curlew	*Numenius arquata*					3w				*	*
Spotted Redshank	*Tringa erythropus*								*		
Common Redshank	*Tringa totanus*		*c			2					*
Green Sandpiper	*Tringa ochropus*						*				
Wood Sandpiper[1]	*Tringa glareola*					3	*				
Turnstone	*Arenaria interpres*										*
Great Skua	*Catharacta skua*							*		*	
Mediterranean Gull[1]	*Larus melanocephalus*						*				
Black-headed Gull	*Larus ridibundus*		*					*			
Common Gull	*Larus canus*		*			2		*			
Lesser Black-backed Gull	*Larus fuscus*							*		*	
Herring Gull	*Larus argentatus*		*d					*			
Kittiwake	*Rissa tridactyla*							*			
Sandwich Tern[1]	*Sterna sandvicensis*					2		*			
Arctic Tern[1]	*Sterna paradisaea*		*								
Little Tern[1]	*Sterna albifrons*		*			3		*			
Common Guillemot	*Uria aalge*									*	
Razorbill	*Alca torda*							*		*	
Black Guillemot	*Cepphus grylle*					2					
Atlantic Puffin	*Fratercula arctica*					2		*			
Stock Dove	*Columba oenas*									*	
Common Cuckoo	*Cuculus canorus*		*								
Barn Owl	*Tyto alba*			*		3					
Short-eared Owl[1]	*Asio flammeus*					3					
Common Kingfisher[1]	*Alcedo atthis*					3					
Green Woodpecker	*Picus viridis*					2					
Sand Martin	*Riparia riparia*					3					
Barn Swallow	*Hirundo rustica*					3					
House Martin	*Delichon urbica*		*								
Tree Pipit	*Anthus trivialis*		*c								
Meadow Pipit	*Anthus pratensis*		*c								
Yellow Wagtail	*Motacilla flava*		*								
Grey Wagtail	*Motacilla cinerea*		*								
Hedge Accentor	*Prunella modularis*		*								
Rufous Nightingale	*Luscinia megarhynchos*			*							
Bluethroat1	*Luscinia svecica*						*				
Black Redstart	*Phoenicurus ochruros*						*				
Common Redstart	*Phoenicurus phoenicurus*					2					
Common Stonechat	*Saxicola torquata*					3					
Fieldfare	*Turdus pilaris*		*				*				
Redwing	*Turdus iliacus*						*				
Mistle Thrush	*Turdus viscivorus*		*								
Dartford Warbler[1]	*Sylvia undata*	*				2		*			
Wood Warbler	*Phylloscopus sibilatrix*		*								
Willow Warbler	*Phylloscopus trochilus*		*								
Goldcrest	*Regulus regulus*		*c								
Firecrest	*Regulus ignicapillus*						*				
Bearded Tit	*Panurus biarmicus*							*			
Golden Oriole	*Oriolus oriolus*						*				
Red-billed Chough[1]	*Pyrrhocorax pyrrhocorax*					3		*			
European Serin	*Serinus serinus*						*				
Lesser Redpoll	*Carduelis cabaret*		*c							*	
Parrot Crossbill	*Loxia pytyopsittacus*						*				
Common Rosefinch	*Carpodacus erythrinus*						*				
Hawfinch	*Coccothraustes coccothraustes*			*							
Snow Bunting	*Plectrophenax nivalis*						*				

Qualifying criteria for Amber listing: HD rec to WI.

1 Annex I of the EC Directive on the Conservation of Wild Birds.
a Subspecies only: European White-fronted Goose *Anser albifrons albifrons*.
b Subspecies only: Greenland White-fronted Goose *Anser albifrons flavirostris*.
c The species is Amber listed, even though its apparent population decline was more than 50%, because this trend is considered to be unrepresentative of the whole of the UK due to geographic or habitat-related sampling bias towards populations with low densities.
d The species is Amber listed, even though its apparent population decline was more than 50%, because the data from Seabird2000 remain provisional.
p The species qualifies as a localised non-breeder for its population occurring on passage.

Table 3. The population status of birds in the UK: 'Green listed' species.

Species	Scientific name	Assessed for BoCCII	SPEC
Little Grebe	*Tachybaptus ruficollis*	Insufficient data	
Great Crested Grebe	*Podiceps cristatus*	✓	
Great Shearwater	*Puffinus gravis*	Insufficient data	
Sooty Shearwater	*Puffinus griseus*	Insufficient data	
Grey Heron	*Ardea cinerea*	✓	
Snow Goose	*Anser caerulescens*	Insufficient data	
Mallard	*Anas platyrhynchos*	✓	
Tufted Duck	*Aythya fuligula*	✓	
Smew[1]	*Mergellus albellus*	✓	3
Red-breasted Merganser	*Mergus serrator*	✓	
Goosander	*Mergus merganser*	✓	
Northern Goshawk	*Accipiter gentilis*	✓	
Eurasian Sparrowhawk	*Accipiter nisus*	✓	
Common Buzzard	*Buteo buteo*	✓	
Hobby	*Falco subbuteo*	✓	
Ptarmigan	*Lagopus mutus*	✓	
Moorhen	*Gallinula chloropus*	✓	
Common Coot	*Fulica atra*	✓	
Little Ringed Plover	*Charadrius dubius*	✓	
European Golden Plover[1]	*Pluvialis apricaria*	✓	4
Sanderling	*Calidris alba*	✓	
Little Stint	*Calidris minuta*	✓	
Curlew Sandpiper	*Calidris ferruginea*	✓	
Jack Snipe	*Lymnocryptes minimus*	✓	3
Greenshank	*Tringa nebularia*	✓	
Common Sandpiper	*Actitis hypoleucos*	✓	
Pomarine Skua	*Stercorarius pomarinus*	Insufficient data	
Arctic Skua	*Stercorarius parasiticus*	✓	
Long-tailed Skua	*Stercorarius longicaudus*	Insufficient data	
Little Gull	*Larus minutus*	✓	3
Iceland Gull	*Larus glaucoides*	Insufficient data	
Glaucous Gull	*Larus hyperboreus*	Insufficient data	
Great Black-backedGull	*Larus marinus*	✓	4
Common Tern[1]	*Sterna hirundo*	✓	
Black Tern1	*Chlidonias niger*	✓	3
Little Auk	*Alle alle*	Insufficient data	
Rock Dove/Feral Pigeon	*Columba livia*	Insufficient data/✓	
Wood Pigeon	*Columba palumbus*	✓	4
Collared Dove	*Streptopelia decaocto*	✓	
Tawny Owl	*Strix aluco*	✓	4
Long-eared Owl	*Asio otus*	✓	
Common Swift	*Apus apus*	✓	
Hoopoe	*Upupa epops*	✓	
Great Spotted Woodpecker	*Dendrocopos major*	✓	
Water Pipit	*Anthus spinoletta*	✓	
Rock Pipit	*Anthus petrosus*	✓	
Pied Wagtail	*Motacilla alba*	✓	
Bohemian Waxwing	*Bombycilla garrulus*	✓	
Dipper	*Cinclus cinclus*	✓	
Wren	*Troglodytes troglodytes*	✓	
Robin	*Erithacus rubecula*	✓	4
Whinchat	*Saxicola rubetra*	✓	4
Northern Wheatear	*Oenanthe oenanthe*	✓	
Blackbird	*Turdus merula*	✓	4
Cetti's Warbler	*Cettia cetti*	✓	
Sedge Warbler	*Acrocephalus schoenobaenus*	✓	4
Reed Warbler	*Acrocephalus scirpaceus*	✓	4
Icterine Warbler	*Hippolais icterina*	✓	4
Lesser Whitethroat	*Sylvia curruca*	✓	
Common Whitethroat	*Sylvia communis*	✓	4
Garden Warbler	*Sylvia borin*	✓	4

Species	Scientific name	Assessed for BoCCII	SPEC
Blackcap	*Sylvia atricapilla*	✓	4
Common Chiffchaff	*Phylloscopus collybita*	✓	
Pied Flycatcher	*Ficedula hypoleuca*	✓	4
Long-tailed Tit	*Aegithalos caudatus*	✓	
Crested Tit	*Parus cristatus*	✓	4
Coal Tit	*Parus ater*	✓	
Blue Tit	*Parus caeruleus*	✓	4
Great Tit	*Parus major*	✓	
European Nuthatch	*Sitta europaea*	✓	
EurasianTreecreeper	*Certhia familiaris*	✓	
Short-toed Treecreeper	*Certhia brachydactyla*	✓	4
Eurasian Jay	*Garrulus glandarius*	✓	
Magpie	*Pica pica*	✓	
Eurasian Jackdaw	*Corvus monedula*	✓	4
Rook	*Corvus frugilegus*	✓	
Carrion Crow	*Corvus corone*	✓	
Common Raven	*Corvus corax*	✓	
Common Chaffinch	*Fringilla coelebs*	✓	4
Brambling	*Fringilla montifringilla*	✓	
Greenfinch	*Carduelis chloris*	✓	4
Goldfinch	*Carduelis carduelis*	✓	
Siskin	*Carduelis spinus*	✓	4
Common Redpoll	*Carduelis flammea*	Insufficient data	
Common Crossbill	*Loxia curvirostra*	✓	
Lapland Longspur	*Calcarius lapponicus*	✓	

SPEC4 = Species of European Conservation Concern which are concentrated in Europe and of favourable conservation status (Tucker & Heath 1994).
1 Annex I of the EC Directive on the Conservation of Wild Birds.

Table 4. Regular breeding species which have become extinct in the UK since 1800. Extinct species are defined as formerly regular breeding birds which have not bred in the UK within the last 20 years.

Species	Scientific name	Scale of extinction	Date last recorded breeding in UK	SPEC category
Great Bustard	*Otis tarda*	Regional	c.1830s	1
Kentish Plover	*Charadrius alexandrinus*	Regional	c.1970s	3
Black Tern	*Chlidonias niger*	Regional	c.1970s	3
Great Auk	*Pinguinus impennis*	Global	c.1800s	–

Bob Glover

263. Marsh Harrier *Circus aeruginosus*. Another species of raptor which has responded well to targeted conservation action, and has moved from the Red to the Amber list.

Table 5. Changes in species composition between BoCC 2002-2007 and the original BoCC list.

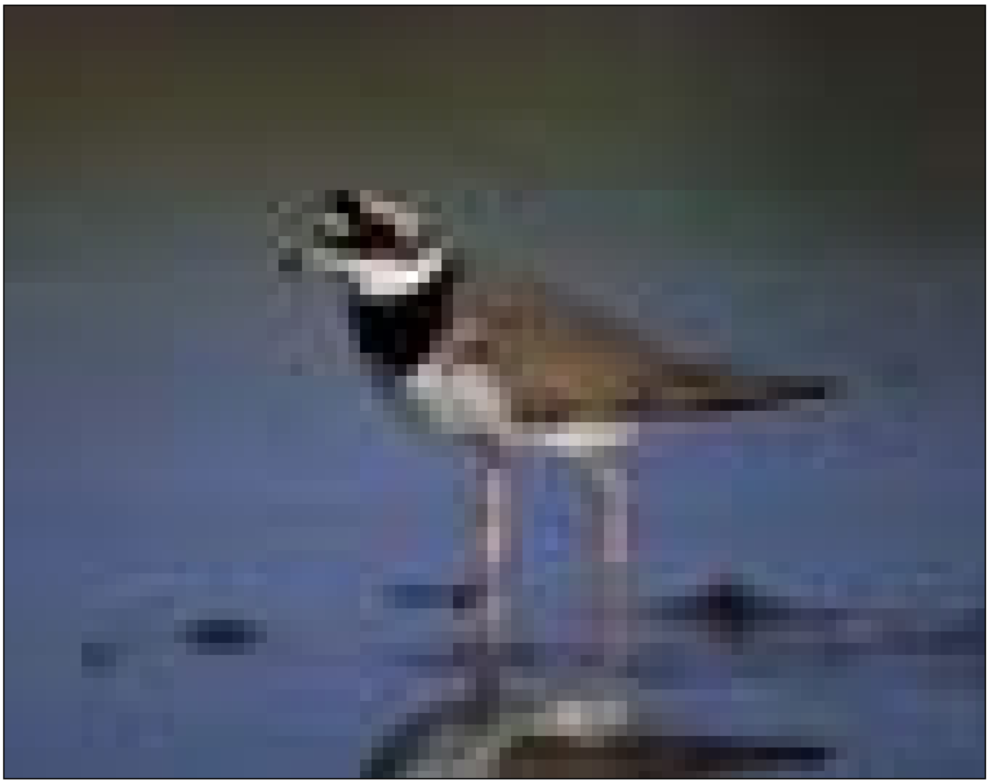
Species	Change in listing	Reason for change in listing
New to Red list		
Lesser Spotted Woodpecker	G fi R	≥ 50% decline in breeding population
Ring Ouzel	A fi R	≥ 50% decline in breeding population
Grasshopper Warbler	A fi R	≥ 50% decline in breeding population
Savi’s Warbler	A fi R	≥ 50% decline in breeding population
Marsh Tit	A fi R	≥ 50% decline in breeding population
Willow Tit	A fi R	≥ 50% decline in breeding population
Common Starling	A fi R	≥ 50% decline in breeding population
House Sparrow	G fi R	≥ 50% decline in breeding population
Yellowhammer	G fi R	≥ 50% decline in breeding population
New to Amber list		
Fulmar	G fi A	Localised during the breeding season
Great Cormorant	G fi A	Localised during the breeding season; internationally important during the non-breeding season
Little Egret	G fi A	Rare breeder
Eurasian Spoonbill	G fi A	Rare breeder
Mute Swan	G fi A	Internationally important during the breeding season
Red Grouse	G fi A	≥ 25% decline in breeding population
Long-tailed Duck	G fi A	Localised during the non-breeding season
Spotted Redshank	G fi A	Localised during the non-breeding season
Green Sandpiper	G fi A	Rare breeder
Black-headed Gull	G fi A	≥ 25% decline in breeding population
Kittiwake	G fi A	Localised during the breeding season
Common Cuckoo	G fi A	≥ 25% decline in breeding population
House Martin	G fi A	≥ 25% decline in breeding population
Tree Pipit	G fi A	≥ 25% decline in breeding population
Meadow Pipit	G fi A	≥ 25% decline in breeding population
Yellow Wagtail	G fi A	≥ 25% decline in breeding population
Grey Wagtail	G fi A	≥ 25% decline in breeding population
Bluethroat	G fi A	Rare breeder
Mistle Thrush	G fi A	≥ 25% decline in breeding population
Wood Warbler	G fi A	≥ 25% decline in breeding population
Willow Warbler	G fi A	≥ 25% decline in breeding population
Goldcrest	G fi A	≥ 25% decline in breeding population
Lesser Redpoll	G fi A	≥ 25% decline in breeding population
From Red to Amber list		
Red Kite	R fi A	≥ 100% increase in breeding population over previous 25 years
Marsh Harrier	R fi A	≥ 100% increase in breeding population over previous 25 years
Osprey	R fi A	≥ 100% increase in breeding population over previous 25 years
Merlin	R fi A	≥ 100% increase in breeding population over previous 25 years
Dartford Warbler	R fi A	≥ 100% increase in breeding population over previous 25 years
From Amber to Green list		
European Golden Plover	A fi G	New data suggest that it does not qualify as internationally important during the non-breeding season
Greenshank	A fi G	New data suggest it does not qualify as a localised breeder
Little Gull	A fi G	No longer a rare breeder (< 1 pair)
Blackbird	A fi G	Population decline < 25% over previous 25 years
Cetti’s Warbler	A fi G	No longer a rare breeder (> 300 pairs)
Icterine Warbler	A fi G	No longer a rare breeder (< 1 pair)
Crested Tit	A fi G	New data suggest it does not qualify as a localised breeder
Brambling	A fi G	No longer a rare breeder (< 1 pair)
Goldfinch	A fi G	Population decline < 25% over previous 25 years
Removed from Amber list		
Black-winged Stilt	A fi Off	No longer a rare breeder (< 1 pair), and excluded because its status as a RBBP species is based on a single summering individual

The lists are abbreviated to: A = Amber; G = Green; R = Red.
For original BoCC list, see Gibbons *et al*. 1996a; JNCC 1996.

Table 6. Changes in the criteria under which Red- and Amber-listed species qualify: new declining species.

Species	Overall listing	Magnitude and season of population decline
Red-throated Diver	Amber	≥ 25% decline in breeding population
Slavonian Grebe	Amber	≥ 25% decline in breeding population
Capercaillie	Red	≥ 50% decline in breeding population
Great Ringed Plover	Amber	≥ 25% decline in non-breeding population
Northern Lapwing	Amber	≥ 25% decline in breeding population
Dunlin	Amber	≥ 25% decline in non-breeding population
Common Snipe1	Amber	≥ 25% decline in breeding population
Woodcock[1]	Amber	≥ 25% decline in breeding population
Black-tailed Godwit	Amber	≥ 25% decline in breeding population
Common Redshank[1]	Amber	≥ 25% decline in breeding population
Common Gull	Amber	≥ 25% decline in breeding population
Little Tern	Amber	≥ 25% decline in breeding population
Fieldfare	Amber	≥ 25% decline in breeding population

1. The species is Amber listed, even though its apparent population decline was more than 50%, because this trend is considered to be unrepresentative of the whole of the UK due to geographic or habitat-related sampling bias towards populations with low densities.



Malcolm Hunt

264. Great Ringed Plover *Charadrius hiaticula*. One of two species of wader (the other is Dunlin *Calidris alpina*) which are now Amber-listed because of declining populations during the non-breeding season.

Table 7. Population trends of species on the Red and Amber lists.

Species	Season	Data source	Region	Period ($t_1 – t_2$)	No. Years	Population t_1	Population t_2	% Change	Caveats	Listing on this criterion
Red-throated Diver	B	Single-species survey[1]	Shetland	1983 – 1994	11	607	390	-36		Amber
Slavonian Grebe	B	Single-species survey[2]	UK	1972/76 – 1997/01	25	61.6	40	-35		Amber
Great Bittern	B	RBBP & Survey[2]	UK	1973/77 – 1997/01	24	50.6	19	-62		Red
Common Scoter	B	RBBP & Survey[3]	UK	1973/77 – 1995	20	149.8	95	-37		Amber
Red Kite	B	RBBP & Survey[4]	UK	1973/77 – 2000	25	32	430	+1244		Amber[a]
Marsh Harrier	B	RBBP	UK	1973/77 – 1995/99	22	8.6	141.4	+1544		Amber[a]
Osprey	B	RBBP	UK	1973/77 – 1995/99	22	15.6	118.2	+658		Amber[a]
Common Kestrel	B	CBC	UK	1974 – 1999	25			-28		Amber
Merlin	B	Single-species survey[5]	UK	1983/84 – 1993/94	10	600	1300	+117		Amber[a]
Red Grouse	B	Single-species survey[6]	UK	1974 – 1999	25			-25 to -49		Amber
Black Grouse	B	Single-species survey[7]	UK	1991/92 – 1995/96	4	25000	6510	-74		Red
Capercaillie	B	Single-species survey[8]	UK	1992/94 – 1998/99	5	2200	1073	-51		Red
Grey Partridge	B	CBC	UK	1974 – 1999	25			-84		Red
Great Ringed Plover	NB	WeBS	UK	1974/75 – 1999/00	25			-28		Amber
Northern Lapwing	B	CBC	UK	1974 – 1999	25			-41	U-H, U-LD	Amber
Northern Lapwing	B	BBS	UK	1994 – 2000	6			-13		
Dunlin	NB	WeBS	UK	1974/75 – 1999/00	25			-36		Amber
Common Snipe	B	CBC	UK	1974–1999	25			-67	SS-L, U-H	Amber[b]
Woodcock	B	CBC	UK	1974 – 1999	25			-76	SS-L, U-H	Amber[b]
Black-tailed Godwit	B	RBBP	UK	1973/77 – 1995/99	22	69.4	45.2	-35		Amber
Common Redshank	B	CBC	UK	1974 – 1999	25			-63	SS-L, U-H	Amber[b]
Common Redshank	B	WBS	UK	1975 – 1999	24			-44	SS, U-H	Amber
Common Redshank	B	Single-species survey[9]	Britain (saltmarsh)	1985 – 1996	11	21022	16433	-23		
Black-headed Gull	B	Seabird trends[10]	UK	1985/88 –1998/01	15	206249	124199	-40		Amber
Common Gull	B	Seabird trends[10]	UK	1985/88 –1998/01	15	67992	42945	-37		Amber
Herring Gull	B	Seabird trends[10]	UK	1969/70 – 1998/01	30	297856	116377	-61		Amber[c]
Roseate Tern	B	Seabird trends[10]	UK	1975	–	2000	25	601	53 -91	Red
Arctic Tern	B	Seabird trends[10]	UK	1985 – 2000	15	80200	44555	-45		Amber
Little Tern	B	Seabird trends[10]	UK	1975 – 2000	25	2644	1902	-28		Amber
Turtle Dove	B	CBC	UK	1974 – 1999	25			-69	SS-L	Red
Common Cuckoo	B	CBC	UK	1974 – 1999	25			-31		Amber

Species	Season	Data source	Region	Period ($t_1 – t_2$)	No. Years	Population t_1	Population t_2	% Change	Caveats	Listing on this criterion
Wryneck	B	RBBP	UK	1973/77 – 1995/99	22	9	3.2	-64		Red
Lesser Spotted Woodpecker	B	CBC	UK	1974 – 1999	25			-73	SS-L	Red
Sky Lark	B	CBC	UK	1974 – 1999	25			-55		Red
House Martin	B	CBC	UK	1974 – 1999	25			-33	SS	Amber
Tree Pipit	B	CBC	UK	1974 – 1999	25			-75	U-H, U-LD	Amber[b]
Meadow Pipit	B	CBC	UK	1974 – 1999	25			-43	U-H, U-LD	Amber[b]
Yellow Wagtail	B	CBC	UK	1974 – 1999	25			-36	SS-L	Amber
Grey Wagtail	B	WBS	UK	1975 – 1999	24			-41		Amber
Hedge Accentor	B	CBC	UK	1974 – 1999	25			-43		Amber
Ring Ouzel	B	Atlas & Survey[11]	UK	1988/91 – 1999	10			-58		Red[d]
Fieldfare	B	RBBP	UK	1973/77 – 1995/99	22	7.6	4	-47		Amber
Song Thrush	B	CBC	UK	1974 – 1999	25			-53		Red
Mistle Thrush	B	CBC	UK	1974 – 1999	25			-38		Amber
Grasshopper Warbler	B	CBC	UK	1974 – 1999	25			-79	SS	Red
Savi's Warbler	B	RBBP	UK	1973/77 – 1995/99	22	11.8	4.4	-63		Red
Marsh Warbler	B	RBBP	UK	1973/77 – 1995/99	22	75	27.6	-63		Red
Dartford Warbler	B	Single-species survey[12]	UK	1974 – 1994	20	560	1670	+198		Amber[a]
Wood Warbler	B	BBS	UK	1994 – 2000	6			-43		Amber
Willow Warbler	B	CBC	UK	1974 – 1999	25			-31		Amber
Goldcrest	B	CBC	UK	1974 – 1999	25			-55	U-H	Amber[b]
Spotted Flycatcher	B	CBC	UK	1974 – 1999	25			-75		Red
Marsh Tit	B	CBC	UK	1974 – 1999	25			-50		Red
Willow Tit	B	CBC	UK	1974 – 1999	25			-80	SS-L	Red
Red-backed Shrike	B	RBBP	UK	1973/77 – 1995/99	22	49	4.8	-90		Red
Common Starling	B	CBC	UK	1974 – 1999	25			-66		Red
House Sparrow	B	CBC[13]	UK	1977 – 1999	22			-62		Red
Tree Sparrow	B	CBC	UK	1974 – 1999	25			-95	SS-L	Red
Linnet	B	CBC	UK	1974 – 1999	25			-55		Red
Lesser Redpoll	B	CBC	UK	1974 – 1999	25			-96	SS, U-H, U-LD	Amber[b]
Bullfinch	B	CBC	UK	1974 – 1999	25			-57		Red
Yellowhammer	B	CBC	UK	1974 – 1999	25			-54		Red
Reed Bunting	B	CBC	UK	1974 – 1999	25			-62		Red
Corn Bunting	B	CBC	UK	1974 – 1999	25			-89	SS-L	Red

Table 7 cont'd. Population trends of species on the Red and Amber lists – KEY.

Species with a population decline ≥ 50% are Red listed, and those with a decline 25.0-49.9% are Amber listed. Seasons are abbreviated to: B = Breeding; NB = Non-breeding. Data sources are abbreviated as follows: BBS = Breeding Bird Survey; CBC = Common Birds Census; RBBP = Rare Breeding Birds Panel; WBS = Waterways Birds Survey; WeBS = Wetland Bird Survey. Population sizes based on RBBP data are five-year means of the maximum total number of pairs (or more relevant unit). The data caveats indicating the reliability of the trends are described in more detail in the text, and are abbreviated in the table as: SS(-E/L) = Small sample size (in early or later years of the time period); U-H/LD = Unrepresentative of the whole of the UK due to habitat- or geographical-related sampling bias towards populations with low densities (Atlas correction ratio ≥ 1).

a Species is Amber listed, even though it has undergone a historical population decline during 1800-1995, because its population size has more than doubled over the previous 25 years.
b The species is Amber listed, even though its apparent population decline was more than 50%, because this trend is considered to be unrepresentative of the whole of the UK due to geographical- or habitat-related sampling bias towards populations with low densities. See text for details.
c The species is Amber listed, even though its apparent population decline was more than 50%, because the data from Seabird 2000 remain provisional.
d Population change incorporates correction for differences in survey effort (Sim, Gregory & Wotton, unpublished data).

1 Gomersall *et al.* (1984); Gibbons *et al.* (1997).
2 RSPB unpublished monitoring data.
3 Underhill *et al.* (1998).
4 Wotton *et al.* (in press).
5 Bibby & Nattrass (1986); Rebecca & Bainbridge (1998).
6 N. J. Aebischer *in litt*.
7 Baines & Hudson (1995); Hancock *et al.* (1999).
8 Catt *et al.* (1998); Wilkinson *et al.* (2002).
9 Brindley *et al.* (1998).
10 Seabird 2000 (Mitchell *et al.* in press)
11 Wotton *et al.* (2002).
12 Bibby & Tubbs (1975); Gibbons & Wotton (1996).
13 The House Sparrow was not reliably recorded on CBC plots before 1976; thus trends are based on the 22-year period 1977-1999.

Mike Lane

265. Tree Sparrow *Passer montanus*. One of a number of farmland birds that remains on the Red list, with its breeding population continuing to decline.

Table 8. Changes in geographical breeding range of species on the Red and Amber lists.

Species this criterion	Data source	Region	Period ($t_1 - t_2$)	No. Years	No. 10-km sq t_1	No. 10-km sq t_2	% Change	Listing on
Great Bittern	Atlas & Survey[1]	UK	1968/72 – 2000	30	35	15	-57	Red
Common Scoter	Atlas & Survey[2]	UK	1968/72 – 1995	25	48	23	-52	Red
Montagu's Harrier	Atlases	UK	1968/72 – 1988/91	20	50	32	-36	Amber
Black Grouse	Atlases	UK	1968/72 – 1988/91	20	603	432	-28	Amber
Capercaillie	Atlases	UK	1968/72 – 1988/91	20	182	62	-64	Red
Water Rail	Atlases	UK	1968/72 – 1988/91	20	707	475	-33	Amber
Corn Crake[a]	Atlas & Survey[3]	Britain	1968/72 – 1998	28	450	93	-79	Red
Stone-curlew	Atlas & Survey[1]	UK	1968/72 – 1996/00	28	93	41	-56	Red
Woodcock	Atlases	UK	1968/72 – 1988/91	20	1814	1259	-31	Amber
Roseate Tern[a]	Atlas & Survey[1]	UK	1968/72 – 2000	30	31	8	-74	Red
Barn Owl	Atlases	UK	1968/72 – 1988/91	20	1884	1151	-39	Amber
European Nightjar	Atlas & Survey[4]	Britain	1968/72 – 1992	22	562	268	-52	Red
Wryneck	Atlases	UK	1968/72 – 1988/91	20	48	6	-87	Red
Wood Lark	Atlas & Survey[5]	UK	1968/72 – 1997	27	196	90	-54	Red
Rufous Nightingale	Atlas & Survey[6]	UK	1968/72 – 1999	29	639	348	-45	Amber
Ring Ouzel	Atlases	UK	1968/72 – 1988/91	20	752	459	-27	Amber
Grasshopper Warbler	Atlases	UK	1968/72 – 1988/91	20	2018	1273	-37	Amber
Marsh Warbler	Atlases	UK	1968/72 – 1988/91	20	21	15	-29	Amber
Red-backed Shrike	Atlases	UK	1968/72 – 1988/91	20	111	15	-86	Red
Hawfinch	Atlases	UK	1968/72 – 1988/91	20	459	315	-31	Amber
Cirl Bunting	Atlas & Survey[7]	UK	1968/72 – 1998	28	174	21	-88	Red
Corn Bunting	Atlases	UK	1968/72 – 1988/91	20	1369	922	-33	Amber

Species with a range contraction ≥ 50% are Red listed, and those with a range contraction 25.0-49.9% are Amber listed. Data sources are abbreviated as follows: Atlases = 1968-72 *Breeding Atlas* (Sharrock 1976) and 1988-91 *New Atlas* (Gibbons *et al.* 1993); Survey = Single-species survey or data gathered as part of an annual monitoring programme.

a Range change based on the number of occupied 10-km squares with breeding evidence only. See text for details.
1 RSPB unpublished monitoring data.
2 Underhill *et al.* (1998).
3 Green & Gibbons (2000).
4 Morris *et al.* (1994).
5 Wotton & Gillings (2000).
6 Wilson *et al.* (in press).
7 Wotton *et al.* (2000).

Table 9. Population estimates for rare breeding species.

Species	Data source	Unit	Population size
Black-throated Diver	Survey (1994)[1]	Pairs	155-189
Red-necked Grebe	RBBP	Max total pairs	2.6
Slavonian Grebe	Survey (1997-2001)[1]	Breeding pairs	40
Black-necked Grebe	RBBP	Max total pairs	55.4
Great Bittern	Survey (1997-2001)[1]	Booming males	19
Little Egret	RBBP	Max total pairs	16.2
Eurasian Spoonbill	RBBP	Max total pairs	3.2
Whooper Swan	RBBP	Max total wild pairs	6.6
Pintail	RBBP	Max total pairs	43.6
Garganey	RBBP	Max total pairs	115.2
Greater Scaup	RBBP	Max total pairs	1.4
Common Scoter	Survey (1995)[2]	Pairs	95
Common Goldeneye	RBBP	Max total pairs	87.8[a]
European Honey-buzzard	Survey (2000)[3]	Max total pairs	61
White-tailed Eagle	Survey (1997-2001)[1]	Territorial pairs	19.6
Marsh Harrier	RBBP	Max breeding females	141.4
Montagu's Harrier	RBBP	Max total pairs	10.6
Osprey	RBBP	Max total pairs	118.2
Spotted Crake	Survey (1999)[4]	Singing males	73
Common Crane	RBBP	Pairs	3.8
Stone-curlew	Survey (2000)[1]	Pairs	254
Temminck's Stint	RBBP	Max total pairs	2.8
Purple Sandpiper	RBBP	Max total pairs	2
Ruff	RBBP	No. leks	3.2
Green Sandpiper	RBBP	Max total pairs	1.4
Black-tailed Godwit	RBBP	Max total pairs	45.2
Wood Sandpiper	RBBP	Max total pairs	8.8
Red-necked Phalarope	Survey (1997-2001)[1]	Males	22.8
Mediterranean Gull	Seabird (1999-2001)	Max total pairs	89
Roseate Tern	Seabird (2001)	Breeding pairs	58
Wryneck	RBBP	Max total pairs	3.2
Bluethroat	RBBP	Max total pairs	1.4
Black Redstart	RBBP	Max total pairs	77.4
Fieldfare	RBBP	Max total pairs	4
Redwing	RBBP	Max total pairs	23.2
Savi's Warbler	RBBP	Max total pairs	4.4
Marsh Warbler	RBBP	Max total pairs	27.6
Firecrest	RBBP	Max total pairs	65.8
Golden Oriole	RBBP	Max total pairs	25.4
Red-backed Shrike	RBBP	Max total pairs	4.8
European Serin	RBBP	Max total pairs (UK)	1.8
Parrot Crossbill	RBBP	Max total pairs	1.8[b]
Common Rosefinch	RBBP	Max total pairs	7.2
Snow Bunting	RBBP	Max total pairs	25.4

Data sources are abbreviated to: RBBP = Rare Breeding Birds Panel; Survey = Single-species survey or data gathered as part of an annual monitoring programme; Seabird = Seabird 2000. The year of the survey is given in parentheses. Population sizes derived from RBBP data are means for the five-year period 1995-99.

a National population figures are no longer collected because only sample data are available from nestbox schemes in Highland Region, Scotland.
b National population status is uncertain.
1 RSPB unpublished monitoring data.
2 Underhill *et al.* (1998).
3 Batten (2001).
4 Gilbert (2002).

P. Newman

266. Hen Harrier *Circus cyaneus*, a Redlist species still in trouble.

Bob Glover

267. The Grey Partridge *Perdix perdix* is another farmland species which, with a decreasing breeding population, remains Red-listed.

Table 10. Localised species during the breeding or non-breeding season.

Species	Season	UK population[1,2,3]	Population on top ten IBAs or SPAs	Percentage of UK population
Red-necked Grebe	NB	200	126	63.0
Fulmar	B	474000[4]	274200	57.9
Manx Shearwater	B	250000	226350	90.5
European Storm-petrel	B	25500[4]	35160	100
Leach's Storm-petrel	B	49000[4]	55000	100
Northern Gannet	B	201000	205870	100
Great Cormorant	B	7700[4]	3900	50.7
Shag	B	26000[4]	13295	51.1
Great Bittern	NB	100	50	50.0
Tundra Swan	NB	8240	8650	100
Whooper Swan	NB	9380	6060	64.6
Bean Goose	NB	400	212	53.0
Pink-footed Goose	NB	241010	254500	100
White-fronted Goose	NB	26840	15265	56.9
Greylag Goose	B	700	425	70.8
Greylag Goose	NB	92610	83450	90.1
Barnacle Goose	NB	67000	47852	71.4
Brent Goose	NB	118740	117590	99.0
Common Shelduck	NB	81890	60360	73.7
Eurasian Wigeon	NB	416740	210400	50.5
Gadwall	B	790	443	56.1
Pintail	NB	28180	24140	85.7
Greater Scaup	NB	12210	9010	73.8
Common Eider	NB	74020	43980	59.4
Long-tailed Duck	NB	16020	8890	55.5
Common Scoter	NB	50150	73150	100
Velvet Scoter	NB	3000	2240	74.7
Common Goldeneye	NB	33220	18155	54.6
Corn Crake	B	648[5]	444	68.5
Oystercatcher	NB	334600	217600	65.0
Avocet	B	492	517	100
Avocet	NB	3395	2755	81.1
Avocet	NB (P)	1760	1735	98.6
Dotterel	B	630[6]	469	74.4
Grey Plover	NB	53250	45050	84.6
Grey Plover	NB (P)	70000	43670	62.4
Red Knot	NB	294900	353800	100
Dunlin	B	9900	6812	68.8
Dunlin	NB	576400	399100	69.2
Ruff	NB	1255	731	58.2
Black-tailed Godwit	NB	15830	13815	87.3
Black-tailed Godwit	NB (P)	12400	14080	100
Bar-tailed Godwit	NB	65430	53260	81.4
Whimbrel	B	530	336	63.4
Whimbrel	NB (P)	3600	4146	100
Spotted Redshank	NB	159	83	52.2
Great Skua	B	8500	6875	80.9
Black-headed Gull	B	124000[4]	62010	50.0
Common Gull	B	43000[4]	34165	79.5
Lesser Black-backed Gull	B	99000[4]	88633	89.5
Herring Gull	B	116000[4]	62820	54.2
Kittiwake	B	360000[4]	324300	90.1
Sandwich Tern	B	12000[4]	10485	85.9
Little Tern	B	1900[4]	1219	64.2
Razorbill	B	120000[4]	70140	58.5
Atlantic Puffin	B	451500	398600	88.3
European Nightjar	B	3400	2021	59.4

Wood Lark	B	1552[7]	1102	71.0
Aquatic Warbler	NB (P)	67	47	70.1
Dartford Warbler	B	1890	1691	89.5
Bearded Tit	B	408	363	89.0
Red-billed Chough	B	342	180	52.6

A site is defined as either an Important Bird Area (IBA), or a Special Protection Area (SPA). Seasons are abbreviated to: B = Breeding and NB = Non-breeding; P = Passage population only. For those species where the UK population estimates were given as ranges, the maximum figure only is presented here. Similarly, the percentage value is the percentage of the UK population maximum. For some species the population occurring on the ten best sites is greater than its UK population estimate because the UK population estimates and IBA or SPA count data originate from different years.

1 Breeding season: BirdLife International/EBCC (2000).
2 Non-breeding season: wildfowl, Kershaw & Cranswick (in prep.) plus WeBS 5-year means for Northern Ireland; waders, Rehfisch *et al.* (in prep.).
3 Passage populations: Stone *et al.* (1997).
4 Seabird 2000 (Mitchell *et al.* in prep.])
5 RSPB unpublished monitoring data.
6 Whitfield (in press).
7 Wotton & Gillings (2000).

Chris Knights

268. The House Sparrow *Passer domesticus* is still a relatively widespread and common species, but is now on the Red list because of population declines. Along with Lesser Spotted Woodpecker *Dendrocopos minor* and Yellowhammer *Emberiza citrinella*, it has moved straight from the Green to the Red list.

Table 11. Species occurring in internationally important numbers during the breeding or non-breeding season.

Species	Season	European population[1,2]	UK population[1,3]	Percentage occurring in UK
Great Northern Diver	NB	5000	3020	60.4
Manx Shearwater	B	340000	250000	74.3
Leach's Storm-petrel	B	152000[4]	49000[4]	32.2
Northern Gannet	B	280000	201000	72.8
Great Cormorant	NB	120000	24900	20.8
Shag	B	80000[4]	26000[4]	33.1
Mute Swan	B	69000	15000	21.7
Tundra Swan	NB	17000	8240	48.5
Whooper Swan	NB	16000	9380	58.6
Pink-footed Goose	NB	225000	241010	100.0
Greylag Goose	NB	105250	92610	88.0
Barnacle Goose	NB	44000	67000	100.0
Brent Goose	NB	325000	138710	42.7
Common Shelduck	NB	300000	81890	27.3
Eurasian Wigeon	NB	1250000	416740	33.3
Gadwall	NB	30000	17350	57.8
Eurasian Teal	NB	400000	197080	49.3
Pintail	NB	60000	28180	47.0
Northern Shoveler	NB	40000	15020	37.6
Common Pochard	NB	350000	84350	24.1
Oystercatcher	B	380000	107000[5]	28.5
Oystercatcher	NB	874000	334600	38.3
Great Ringed Plover	NB	47500	32980	69.4
Grey Plover	NB	168000	53250	31.7
Northern Lapwing	NB	7000000	2079000	29.7
Red Knot	NB	345000	294900	85.5
Purple Sandpiper	NB	50500	17740	35.1
Dunlin	NB	1373000	576400	42.0
Black-tailed Godwit	NB	65000	15830	24.4
Bar-tailed Godwit	NB	115000	65430	56.9
Eurasian Curlew	B	330000	100000[5]	30.2
Eurasian Curlew	NB	348000	155700	44.7
Common Redshank	NB	177000	125500	70.9
Turnstone	NB	67000	51960	77.6
Great Skua	B	15000	8500	58.1
Lesser Black-backed Gull	B	270000[4]	99000[4]	37.3
Common Guillemot	B	2400000[4]	930000[4]	39.2
Razorbill	B	580000[4]	120000[4]	20.7
Stock Dove	B	650000	240000	36.7
Scottish Crossbill	B	1250	1250	100.0

Seasons are abbreviated to: B = Breeding; NB = Non-breeding. For those species where the European and/or UK population estimates were given as ranges, the maximum figure only is presented here. Similarly, the percentage value is the percentage of the European population maximum.

1 Estimates of European and UK breeding populations: BirdLife International/EBCC (2000)
2 Estimates of European non-breeding populations: Rose & Scott (1997) northwest European population (wildfowl and allies): Smit & Piersma (1989) East Atlantic Flyway population (waders).
3 Estimates of UK non-breeding populations: Kershaw & Cranswick (in prep.) plus WeBS 5-year means for Northern Ireland (wildfowl); Rehfisch *et al.* (in prep.) (waders).
4 Updated by Seabird 2000 (Mitchell *et al.* in prep.).
5 O'Brien *in litt.*

Table 12. Population trends, localness and international importance of geese (species and subspecies/races/populations) occurring in the UK.

Species	Subspecies/race/population	Scientific name	Season	Population change (%)[1]	Percentage on top ten IBAs or SPAs	NW European population[2]	UK population[3]	Percentage occurring in UK
Bean Goose		*Anser fabalis*						
	Taiga Bean Goose	*Anser fabalis fabalis*	NB		53.0	80000	400	0.5
White-fronted Goose		*Anser albifrons*	NB		56.9	630000	26840	4.3
	Greenland White-fronted Goose	*Anser albifrons flavirostris*	NB		56.1[b]	30000	21050	70.2[d]
	European White-fronted Goose	*Anser albifrons albifrons*	NB	-27[a]	93.0[b]	600000	5790	1.0
Greylag Goose		*Anser anser*	NB		90.1	105250	92610	88.0
	Icelandic Greylag Goose	*Anser anser (Icelandic)*	NB		100[b]	100000	82990	83.0[d]
	Hebridean Greylag Goose	*Anser anser (Hebridean)*	B		70.8[c]			
	Hebridean Greylag Goose	*Anser anser (Hebridean)*	NB		5250	9620	100d	
Barnacle Goose		*Branta leucopsis*	NB		71.4	44000	67000	100
	Greenland Barnacle Goose	*Branta leucopsis (Greenland)*	NB		79.4[b]	32000	45000	100[d]
	Svalbard Barnacle Goose	*Branta leucopsis (Svalbard)*	NB		58.3[b]	12000	22000	100[d]
Brent Goose		*Branta bernicla*	NB		99.0	325000	138710	42.7
	Dark-bellied Brent Goose	*Branta bernicla bernicla*	NB		100[b]	300000	98110	32.7[d]
	Svalbard Light-bellied Brent Goose	*Branta bernicla hrota (Svalbard)*	NB		71.7[b]	5000	2900	58.0[d]
	East Canadian Light-bellied Brent Goose	*Branta bernicla hrota (East Canadian)*	NB		89.6[b]	20000	17730	88.7[d]

1 Population change is for the 25-year period 1974/75-1999/2000.
2 NW European population estimates taken from Rose & Scott (1997).
3 UK population estimates calculated using Kershaw & Cranswick (in prep.) and WeBS 5-year means for Northern Ireland.
a Subspecies would qualify on the Amber list as a declining non-breeder.
b Subspecies would qualify on the Amber list as a localised non-breeder.
c Subspecies would qualify on the Amber list as a localised breeder.
d Subspecies would qualify on the Amber list as internationally important during the non-breeding season.

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Red to the Amber list because their populations have more than doubled in the last 25 years (Red Kite, Marsh Harrier *Circus aeruginosus*, Osprey *Pandion haliaetus*, Merlin *Falco columbarius* and Dartford warbler *Sylvia undata*).

Thirty-one species remain Red listed. A small number of these are listed under different or additional criteria from the original BoCC list because of recent changes in population size or range. The rapid population decline of Common Scoter, for which it was originally Red listed, has lessened so it now qualifies as Amber on this criterion; however, it remains Red listed because it has suffered a rapid contraction in its breeding range. Conversely, Capercaillie *Tetrao urogallus*, originally Red listed by virtue of its range contraction, is now known to be undergoing a rapid population decline. The population decline of the Wryneck *Jynx torquilla*, also originally Red listed for its range contraction, has increased, so it now qualifies on rapid, rather than moderate, population decline. In contrast, the populations of both Corn Crake and Stone-curlew have increased considerably over recent years and no longer qualify on the basis of a declining breeding population, though both remain Red listed because of a contraction in their breeding range. The Corn Crake remains, in addition, Globally Threatened and is a species which has suffered historical decline. The breeding range of Roseate Tern has contracted further and the species now qualifies as a Red-listed, rather than an Amber-listed, species under this criterion (table 8).

Despite the taxonomic uncertainties over the species status of the Scottish Crossbill *Loxia scotica* (Piertney *et al*. 2001), it remains Red listed based on a revised assessment of its global threat status using IUCN criteria. Assuming that it is a full species, it qualifies as Globally Threatened (Vulnerable) because of a restricted global breeding range (extent of occurrence).

Twenty-three species have moved from the Green to the Amber list (table 5). Of these, thirteen species (Red Grouse *Lagopus lagopus*, Black-headed Gull *Larus ridibundus*, Common Cuckoo *Cuculus canorus*, House Martin *Delichon urbica*, Tree Pipit *Anthus trivialis*, Meadow Pipit, Yellow Wagtail *Motacilla flava*, Grey Wagtail, Mistle Thrush *Turdus viscivorus*, Wood Warbler *Phylloscopus sibilatrix*, Willow Warbler *P. trochilus*, Goldcrest *Regulus regulus* and Lesser Redpoll *Carduelis cabaret*) have moved because of declining populations. Even though the measured declines of Common Snipe, Woodcock, Common Redshank, Tree Pipit, Goldcrest and Lesser Redpoll are of more than 50%, they are Amber rather than Red listed here because their population trends may well be unrepresentative due to geographical or habitat-related bias. The Red Grouse is Amber listed based on composite information from game-bag records and local surveys from across the UK. Wood Warbler is Amber listed based on a new (BBS) dataset, while Grey Wagtail moves from Green to Amber because of its WBS trend.

Chris Knights

269. Northern Lapwing *Vanellus vanellus.* This species is on the brink of being Red listed, as a result of steep population declines, especially in England and Wales; however, it remains Amber-listed.

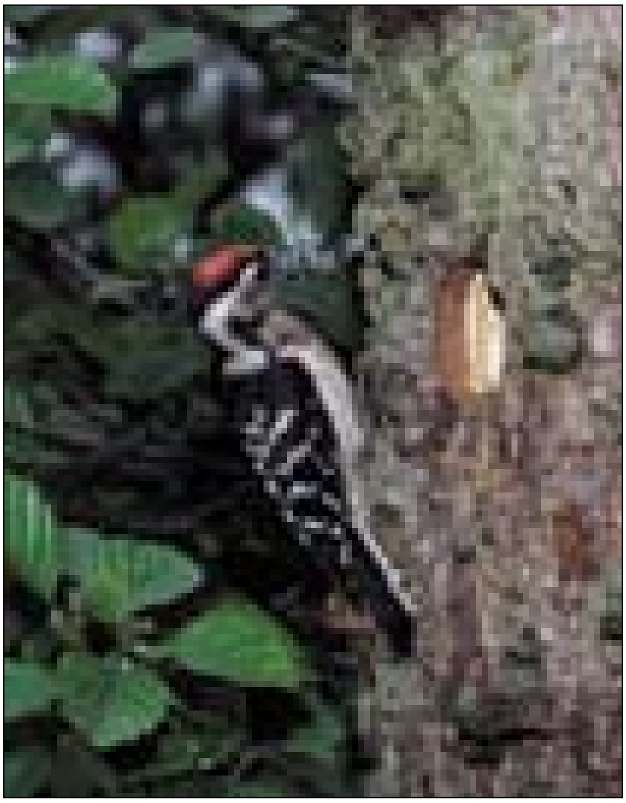
Meadow Pipit is Amber listed because of a population decline; indeed, had unrepresentative trends been allowed for in the original listing, it would have been Amber-listed then. Four species moved from Green to Amber because they now qualify as rare breeders (Little Egret *Egretta garzetta*, Eurasian Spoonbill *Platalea leucorodia*, Green Sandpiper *Tringa ochropus* and Bluethroat *Luscinia svecica*) (table 9). The remaining four species now qualify as localised breeders (Fulmar *Fulmarus glacialis*, Great Cormorant and Kittiwake *Rissa tridactyla*) or non-breeders (Spotted Redshank *Tringa erythropus*) (table 10), or as internationally important during the breeding season (Mute Swan *Cygnus olor*) or non-breeding season (Great Cormorant) (table 11).

As with the Red list, a number of species on the Amber list are listed under different or additional criteria compared with the original BoCC list because of recent changes in population size. Of these, seven species (Red-throated Diver, Slavonian Grebe, Northern Lapwing, Black-tailed Godwit *Limosa limosa*, Common Gull *Larus canus*, Little Tern *Sterna albifrons*, and Fieldfare *Turdus pilaris*) have now all undergone moderate declines in their breeding populations (table 6). The Herring Gull *Larus argentatus* remains on the Amber list, even though it appears to have declined by around 60%, because the data from Seabird 2000 are incomplete at the time of writing. Based on new statistical analyses of CBC trends, Barn Swallow *Hirundo rustica* no longer qualifies as a declining breeder, but it remains on the Amber list because of an unfavourable conservation status in Europe. In addition, two species (Great Ringed Plover *Charadrius hiaticula* and Dunlin *Calidris alpina*) are now Amber listed because of declining populations during the non-breeding season.

Nine species have been moved from Amber to Green (table 5). Blackbird *Turdus merula* and Goldfinch *Carduelis carduelis* have moved off the Amber list because their population declines have lessened, though only marginally so for Blackbird (its decline is now 22%). Little Gull, Cetti's Warbler *Cettia cetti*, Icterine warbler *Hippolais icterina* and Brambling *Fringilla montifringilla* all move to Green because they no longer qualify as rare breeders, Cetti's Warbler because of an increasing population, the other three because they have not bred in recent years. Greenshank *Tringa nebularia* and Crested Tit no longer qualify as localised breeders and European Golden Plover does not qualify as internationally important during the non-breeding season, based on new UK population estimates. In addition, Black-winged Stilt now drops off the BoCC list because its inclusion, as a RBBP species, was based on a single summering individual.

Discussion

This is the third review of the population status of birds in the UK. The analysis uses the most up-to-date information on populations to place species in one of three lists according to a set of



George McCarthy

270. Lesser Spotted Woodpecker *Dendrocopos minor*. The population of this elusive woodland species has decreased sharply, and it is now Red-listed.

scientific criteria. The new listings describe the population status of each species and will, when combined with additional information, help to inform conservation action during the period 2002-2007. The review does not consider legal frameworks for the conservation of species, which would be relevant to conservation action, nor does it look to recreate populations from the very distant past, some of which might be realistic to attain, others of which would not. The overall aim of the paper is to provide information which will help maintain and conserve the diversity of UK bird populations into the future.

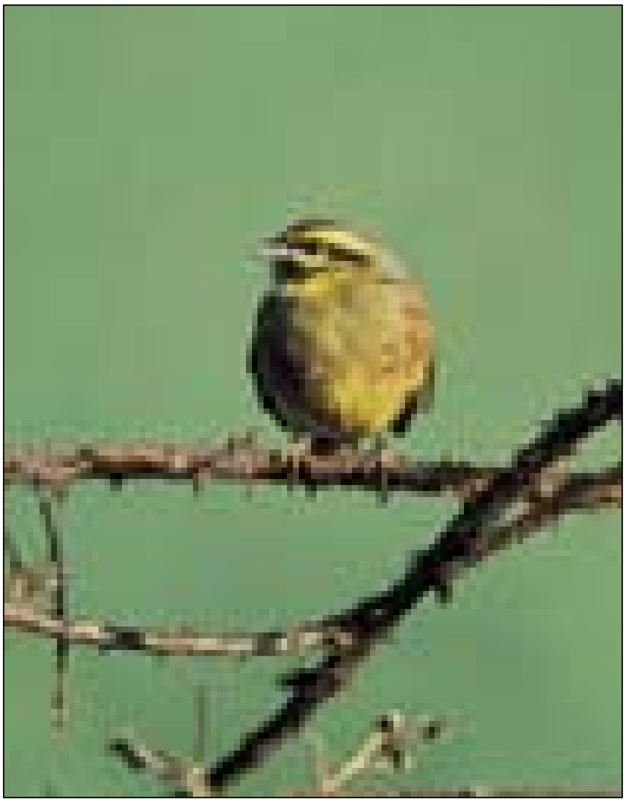
Composition of the new and old lists

The new list provides an insight into the current state of the UK's birds. Effective and targeted conservation action is largely responsible for moving five species from the Red list to the Amber list (Red Kite, Marsh Harrier, Osprey, Merlin and Dartford Warbler), because their populations have more than doubled in the last 25 years. This success is a direct reflection of the considerable efforts made by governmental and non-governmental organisations. We should not, however, lose sight of the fact that, in many cases, the populations of these species remain fragile, well below historic levels, and most are dependent on sustained conservation action.

Nine new species have been added to the Red list because of population decline (Lesser Spotted Woodpecker, Ring Ouzel, Grasshopper Warbler, Savi's Warbler, Marsh Tit, Willow Tit, Common Starling, House Sparrow and Yellowhammer). They represent a relatively varied group of birds but it is notable that several are relatively widespread and common species. Indeed, sixteen species on the Red list have current populations in excess of 10,000 pairs or territories (table 1). In general, conservation policies and actions to aid population recovery will tend to differ for common and rare species. The Yellowhammer is added to the list of rapidly declining farmland species, while the inclusion of Lesser Spotted Woodpecker, Marsh Tit and Willow Tit suggests that some woodland species may be struggling to maintain their populations. According to the CBC, three other woodland species (Woodcock, Tree Pipit and Lesser Redpoll) have also declined by more than 50% over the last 25 years. These trends are, however, unlikely to be representative of their entire UK population, so they have been placed on the Amber list.

There are now two species (Great Ringed Plover and Dunlin) the populations of which have declined by more than 25% over 25 years during the non-breeding season. This is in contrast to the original BoCC list, in which no species qualified by virtue of a declining population in the non-breeding season. The admission of species under this criterion highlights the need for comprehensive monitoring of waterbirds in the non-breeding season.

The new Red list has four more species than the last, and the Amber list has eleven more species. In the vast majority of



Mike Lane

271. Cirl Bunting *Emberiza cirlus* is still Red-listed, because of the contraction of its breeding range, even though it has responded well to recent efforts to stem its decline.

cases, these differences reflect a genuine change in status, particularly for declining populations, rather than any alteration of the criteria used to derive the listings. Inevitably, a number of species come close to Red and Amber listing as declining breeders, but just fail to meet the criteria. The most prominent example on the threshold for Red listing is the Northern Lapwing, which has undergone a sharp population decline in many parts of the UK, particularly in Wales and England (Wilson *et al.* 2001), but overall qualifies under moderate decline for the Amber list. Other species in this category include Grey Wagtail, Hedge Accentor *Prunella modularis* and Fieldfare, all of which have declined by more than 40% in the last 25 years. Research is underway to investigate the reasons for the decline of some of these species (e.g. Northern Lapwing) and we continue to monitor their populations closely. Likewise, a small number of species fall just below the threshold for Amber listing, including Moorhen *Gallinula chloropus* (-21%) and Blackbird (-22%) over 25 years, and Whinchat *Saxicola rubetra* (-21%) over the last six years.

Time windows

This list is designed to inform discussion on the conservation status of birds over the period 2002-2007. It is based, as far as possible, on data from two periods: the last 25 years and the last 200 years (historical decline). The first category identifies current status; the second accounts for a small number of species the current populations of which lie far below their historical levels. Both time windows are to some degree subjective, being driven by the availability of data. For the population decline criterion, the use of the 25-year window introduces an obvious difficulty: as it is rolled on from revision to revision, some species the populations of which have fallen in the recent past will simply disappear from the Red list, even if they have not recovered to any degree. We consider this to be unsatisfactory: if conservation actions have failed to stimulate the recovery of such species towards their former levels then, arguably, they should remain of conservation concern.

In devising the new list, there was much discussion about the need for an intermediate period between the last 25 years and the last 200 years, and we considered using 30-year trends alongside those for 25 years from the CBC. In this instance, there was nothing to be gained by using the longer period because the trends proved to be so similar. Future revisions of the listings may, however, need to supplement the 25-year trends by longer periods (e.g. 35-year trends).

An alternative approach to the same problem is to move a species off the Red list only when it can be shown to have 'recovered'. This would mean that, once Red listed, a species would remain so until it can be shown to have recovered. We considered two ways of defining recovery. In the first, it is suggested that a species can move from Red to Amber only when it can be shown to have doubled in numbers during the previous 25 years. In the second, it is moved to Amber only when it can be shown to have reached half its former population level. Each approach has shortcomings: the first does not work well when populations become very small (less than 20 pairs) because a doubling might, in absolute terms, be relatively trivial, while the second requires an informed decision about an ideal or natural population level. Adoption of either method could increase the length of the Red list in the event that conservation action failed to reverse rapid declines and, by inference, downgrade the importance of more recent trends (i.e. over the last 25 years). This approach would alter the balance of the listing procedure to highlight declines in the past. It should be noted that this is not an issue for this version of the list: no species is removed from the Red list due to the changed period sampled by the 25-year time window. It is for those involved in future BoCC revisions to decide how to tackle such issues.

Absolute change versus rates of change

In this paper, we follow Gibbons *et al* (1996a) in using absolute changes in population size or level over fixed periods, rather than using the rates of change, regardless of the time span. The latter approach is used within the North American listing process (Beissinger *et al.* 2000; Carter *et al.* 2000) and was considered by this review. Some of the arguments for using the rates of change are compelling, since we can convert population trend figures for any period into per annum rates and judge these against our predefined thresholds. Ultimately, it is the rate of decline, i.e. the slope of the trend, in which we are most interested. A 50% decline over 25 years, for example, is equivalent to a 2.7% per annum change; consequently, per annum changes of 2.7% or more could qualify species for the Red list. The same procedure could be used to identify Amber-list species, using a per annum rate of ≥ 1.14%.

One of the drawbacks of this approach, however, is the inherent uncertainty in extrapolating trends from relatively short periods to a nominal 25-year span. For example, a trend over five, or even ten years, may be at a rate equivalent to a 50% reduction over 25 years, but it is very different from a measured trend *throughout* that 25-year period. The shorter the time span, the more uncertain this extrapolation becomes. In fact, from what we know about population fluctuations in birds, it would seem unsafe to assume anything approaching a constant rate of change. The other problem with this approach is that it would depart from the original listing process and the new list would partly reflect changes in the method, rather than changes in the status of birds.

A further complication underlying the listing process is that the halving of a population of, for example, Wrens *Troglodytes troglodytes* might be viewed as of lower conservation concern than the halving of a population of Golden Eagles *Aquila chrysaetos*, because their life histories, and hence their intrinsic ability to recover, are so different. One can imagine ways in which life history could be incorporated into this review, but it is not straightforward. These are likely to be issues for future revisions to consider.

The use of IUCN regional guidelines

The recent publication of regional guidelines for the application of the IUCN Red-list criteria (Gärdenfors 2001; Gärdenfors *et al.* 2001) opens the way for future revisions of BoCC to follow their lead. Greater standardisation of listing criteria would be highly desirable because it would make comparisons between countries more transparent and meaningful. The work by IUCN remains in progress, however, and we considered the guidelines too preliminary to be adopted within this review. The criteria have been used for a range of taxa, including birds, in Sweden, Finland and Switzerland (Gärdenfors 2001), although they are not without problems. It should be borne in mind that the guidelines identify regional extinction probabilities and *not* population status *per se*. As described earlier, the IUCN advises that conservation priorities should consider a range of practical, logistical and political factors in addition to population status. The outcome in Switzerland, for example, is that their new bird list is partially based on the regional guidelines and, by necessity, partially on specific, new criteria (Keller & Bollmann 2001). A potential problem is that if the 'new criteria' differ between countries then the lists become much less comparable. The other major difficulty in applying the regional guidelines is to decide if a 'metapopulation' is a source or a sink (Pulliam 1988; Watkinson & Sutherland 1995). To apply the criteria at a national level, one would first apply the global criteria and then decide whether to upgrade, downgrade or accept the resulting category based on the probability that the population could be 'rescued' by immigrants from neighbouring countries (Gärdenfors 2001; Gärdenfors *et al.* 2001). The problem, even in a country like the UK where good data exist, is that we lack information on the structure of 'metapopulations' and such information is extremely difficult, if not impossible, to collect. The alternative would be to use a surrogate or surrogates of immigration/emigration probability to identify sources and sinks.

Notwithstanding these difficulties, IUCN's draft guidelines should be warmly welcomed as they begin a process of discussion and development which will help to standardise methodologies and terminology in future Red lists. In a separate exercise, we are comparing the listing processes, as applied to birds in the UK, using IUCN's regional guidelines and the criteria in this paper.

Data availability

The monitoring information available for this review is unparalleled when compared with other countries and other taxa. The monitoring data for birds in the UK are far more diverse, more detailed and cover a much longer time span than in most other countries. There are, nonetheless, problems of data availability and quality for some species. For example, a whole group of scarce but widespread breeding species are poorly covered by current and former monitoring schemes, such as Little Grebe, Common Shelduck *Tadorna tadorna*, Common Snipe, Woodcock, Eurasian Curlew, Common Redshank, Hawfinch *Coccothraustes coccothraustes*

Tony Hamblin

272. Willow Tit *Parus montanus*. Like the Marsh Tit *P. palustris*, this is another woodland species which is now on the Red list because its population is in decline.

and Common Crossbill *Loxia curvirostra*. Among the rare breeding birds, there is a group which occurs sporadically and unpredictably over large, often remote areas, e.g. Fieldfare and Redwing *Turdus iliacus*, which makes data collection and interpretation extremely difficult. In addition to the problem of species coverage, there is that of small sample sizes. Since rare breeding birds occur in very low numbers, a small numerical difference from year to year can equate to significant proportional change. We have attempted to overcome this problem by comparing population figures averaged over a number of years. We had also considered, but then dismissed, the idea of setting arbitrary thresholds below which species were excluded from the assessment. The obvious problem is that this would exclude high and low quality data in an indiscriminate manner.

There are three other groups of species where current monitoring information is inadequate: passage seabirds (e.g. shearwaters and skuas), which probably occur in considerable numbers in UK waters; widespread wintering species which occur in terrestrial habitats (e.g. European Golden Plover, Northern Lapwing, Fieldfare and Redwing); and, finally, several waders and waterfowl for which it is extremely difficult to assess the size and/or trends, and hence the importance of their passage populations.

Subspecies

This paper follows previous work (Batten *et al.* 1990; Gibbons *et al.* 1996a; JNCC 1996) in considering only full species, although we present comparable information for certain subspecies or populations of geese (table 12). The latter are something of a special case because some of the international treaties related to their conservation are based on biogeographical populations (e.g. the European Union's Directive on the Conservation of Wild Birds, and the Agreement on the Conservation of African-Eurasian Migratory Waterbirds). While there are various other subspecies present in the UK, the argument to promote their conservation over full species is not compelling and there is uncertainty over the genetic divergence among subspecies and species. Ongoing research by the BOU may well result in some current subspecies taking on full species status and we recommend that future revisions of this list continue to follow the expert advice provided by the BOU.

Introduced species

Introduced species, as defined by the BOU, are excluded from this assessment because their populations are of no conservation concern within the UK; except, that is, for their potentially harmful impact upon native taxa both here and abroad as their populations increase. It was once argued that because the UK supports populations of globally uncommon species, such as Mandarin Duck *Aix galericulata*, Golden Pheasant *Chrysolophus pictus* and Lady Amherst's Pheasant *C. amherstiae*, special attention should be paid to their conservation in this country, but we see no compelling reason to attach conservation concern to these species in a UK context. Indeed, if these species have conservation problems, then the appropriate response is to address the causes of these within their native ranges.

Country-based lists

Devolved governance within the UK raises the possibility of population reviews for individual countries. Such lists need to be hierarchical in nature so that Global, European and UK status are overlaid upon species' status within countries. This is necessary to ensure that the wider conservation status of species is considered within countries. The inherent mobility of birds, and connectivity of populations in the UK, complicates the interpretation of such lists. One of the practical problems in carrying out country reviews is the availability of adequate monitoring data because monitoring programmes are largely UK-based, particularly for more common and widespread species. The most obvious pitfall in drawing up lists at increasingly small spatial scales is that extreme rarities within a region can be elevated in status, even though they may be widespread and abundant elsewhere. On the other hand, conservation action at the edge of a species' range may be important in maintaining the extent of its range. It is also possible for the fate of a non-migratory species in a region to be determined by events entirely outside its borders and beyond its control through emigration and immigration. A further problem is that as areas and sample sizes become smaller, trend analysis becomes more uncertain, since for some species adequate data is completely lacking, and hence species selection is difficult and prone to error. The main implication is that robust regional listings require support for monitoring at the appropriate spatial scale. Nonetheless, lists which are able to target species that have suffered a disproportionate decline within a region or country, and can thus engage and communicate with the relevant decision makers, might serve a useful purpose.

The future

The dynamic nature of bird populations means that numbers and range can alter rapidly over relatively short periods. It is thus appropriate to review the status of bird populations on a regular basis. We recommend that reviews be carried out every five years, and for the next one to be published in 2008.

Acknowledgments

We would like to thank Mark Avery, Stephen Baillie, Helen Baker, Des Callaghan, Roger Clarke, Peter Cranswick, Andy Douse, Keith Duff, Ian Enlander, Ulf Gärdenfors, Rhys Green, Craig Hilton-Taylor, Baz Hughes, Melanie Kershaw, Keith Kirby, Ian McLean, Ian Mitchell, Mark O'Connell, Howard Platt, Jim Reid, Malcolm Smith, David Stroud, Malcolm Vincent, Sian Whitehead and Gwyn Williams.
We are very grateful to the following people for providing information: Mike Austin, Adam Burrows, Len Campbell, Gillian Gilbert, Ian Johnstone, Rob Lucking, John Marchant, Ian Mitchell, Mark O'Brien, Georgina Pickerell, Norman Ratcliffe, Innes Sim, Ken Smith, Ron Summers, Andrew Wilson and Simon Wotton.

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